

人生如戲戲如人生—虛與實之間

人生は芝居のよう; 芝居は人生のよう—
偽と真實の間

Life is like a Game and a
Game is like Life— between
Virtual and Real

1

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Outline

- What are games?
- Some earlier projects on animated interface virtual agents
- Models of agent's minds: BDI, emotion, personality and social interactions
- Interactive drama and story generation using intelligent believable agents
- Dialogue generation in agent-based games
- How can virtual agents become social aware while interacting with human ?

16/5/30

What are games?

- Human life is full of games:
 - Mating is a game
 - Evolution is a game
 - Politics are games
 - War is a game
 - Stock market is a game
 - Competition with environment/territory is a game
 - Survival is a game
 - Sports are games...
- Winner/looser? Cheating or Honest? Right or Wrong? Good or Bad?
- How about virtual worlds?

Multi-characteristics of games

- Games implies competition/cooperation among agents: soccer games; chess; auction (winner/loser)
- Game trees: AND/OR tree, alpha-beta search tree...
- Game theory: explain conflicting decision making among intelligent agents.
 - Solution concept as a Nash equilibrium
 - Prisoner's dilemma

Entertainment games

- Fighting, enthusiastic, education, fun
- participating by human players
- Interactive games: Wii, x-box kinect
- Role playing games, human agents and virtual agents involved in games , Second life
- Interactive Drama: human and virtual players involve in development of a story

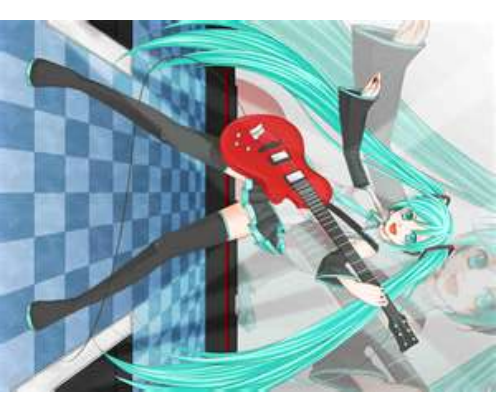


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Agent-based entertainment games applications

- Cartoon/commercial characters:
 - Virtual singers
 - Virtual news reporter
 - Virtual receptionist
 - Virtual creatures
 - Virtual pets, etc.
- User friendly interface: Interacting agents
- Character Agents in Entertainment games
- Education pedagogical / tutorial agents



16/5/30

Issues on agent-based simulation in role player games

16/5/30

- Can virtual agents simulate human players?
Make the agents believable:
 - Similar reasoning and intelligence: understanding problems, situations, risk, infer solutions
 - Creativity and learning
 - Similar emotional responses
 - Similar personality and culture oriented behaviors
 - Similar social reactions/skills
 - Sense of humor...

Modeling of agent's minds, emotions and social interactions

- Agents mind modeling: theory of mind
 - perception, belief, desire, emotion, personality, preference, mood, goals, intention, actions of agents
- Emotions OCC model: appraisal and coping
- Personalities OCEAN model
- Social relations and social norms: permission, obligation, commitments, value systems
 - Social planning agents: interacting and modeling with environments and other agents

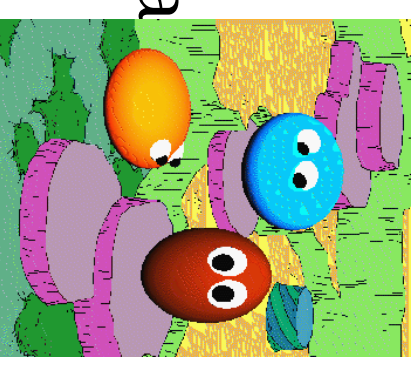
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Issues in human-agent interactions

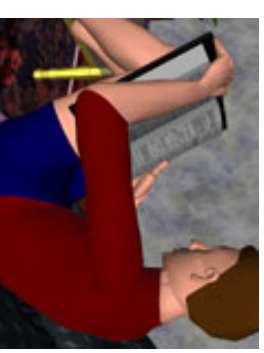
- Animation and humanoid behavior
 - Facial expression, body movement, eye-gaze, gesture
- Multi-modal interaction
 - Gesture, facial expression, text, speech, graph, video
- Natural language interface
 - Speech and dialogue: tone and emotion
 - Natural language texts
- Model of emotional states in terms of the multi-modal agent behaviors parameters
- Individual or collective behaviors: Artificial life—vivid imitation behaviors of fish, birds, etc

Early projects on animated agents

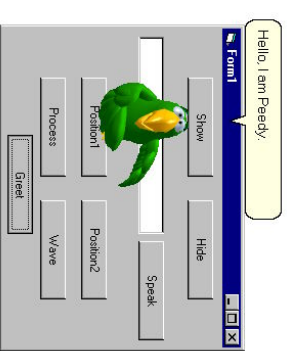
- The Woggles, individual personalities which engage in social behaviors and display emotions in interactive drama (Oz Project, CMU), 1990



- Jack, articulated figure with realistic behavior (HMS, university of Pennsylvania) Human Modeling, 1994



- Peedy, Conversational assistant (Microsoft Research): text to speech 1996



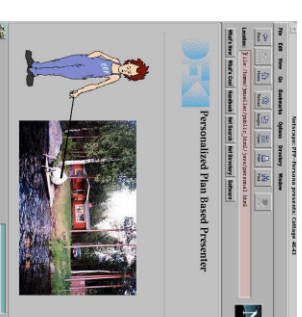
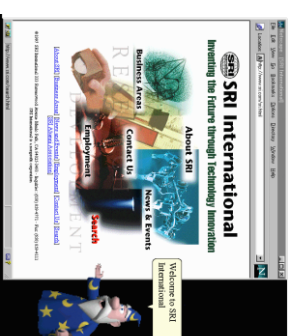
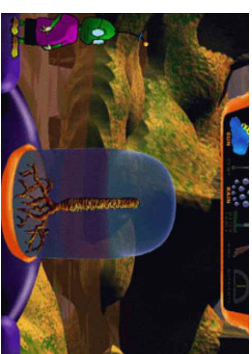
STEVE— A pedagogical agent

- Animatd agent for procedural training in virtual environment 1997
- USC/ISI (Soar Training Expert for Virtual Environments) that supports the learning process.
- Steve agents can demonstrate skills to students, answer student questions, watch the students as they perform the tasks, and give advice if the students run into difficulties.
- Multiple Steve agents can inhabit a virtual environment, along with multiple students. This helps make it possible to train students on team tasks.



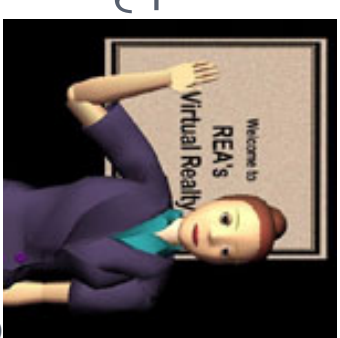
Projects on animated interface agents

- Herman, Pedagogical Agents for Design-A-Plant, 1997 (North-Carolina State U.)
- Cosmo, a life-like Pedagogical Agent for the Internet Advisor System (North Carolina State University) 1997
- Info Wiz: An Animated Voice for Interactive Interface to the Web (SRI International) 1997
- Persona, an animated Presentation Agent of the PPP system (DFKI GmbH) 1996



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MULTIMODAL CONVERSATIONAL HUMANOID --MIT Media lab project

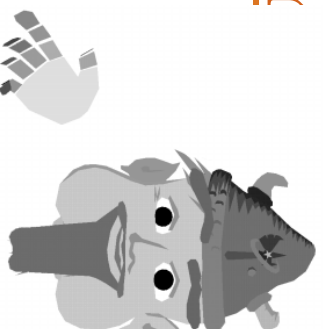


6/5/30

- ☹ REA project: Mimic human real-estate agent to interact with users in natural language. 2000 (dialogue conversation)
- Gandalf project: Solar system experts to talk with humans in natural language and gestures about traveling in planet, 1996

<http://alumni.media.mit.edu/~gandalf.html>

[kris](#)



Early project with emotion agents

- ALIVE: MIT by P. Maes 1993



16/5/30

An Artificial Life Interactive Video Environment

Allow wireless full-body interaction between a human participant and a graphic worlds inhabited with autonomous agents (with **needs** and **motivations**)

- CyberCafe, Virtual Theater: Stanford, KSL, by B. Hayes-Roth -- waiter and customers 1998
(**personality** and **mood**)

Projects about story generation

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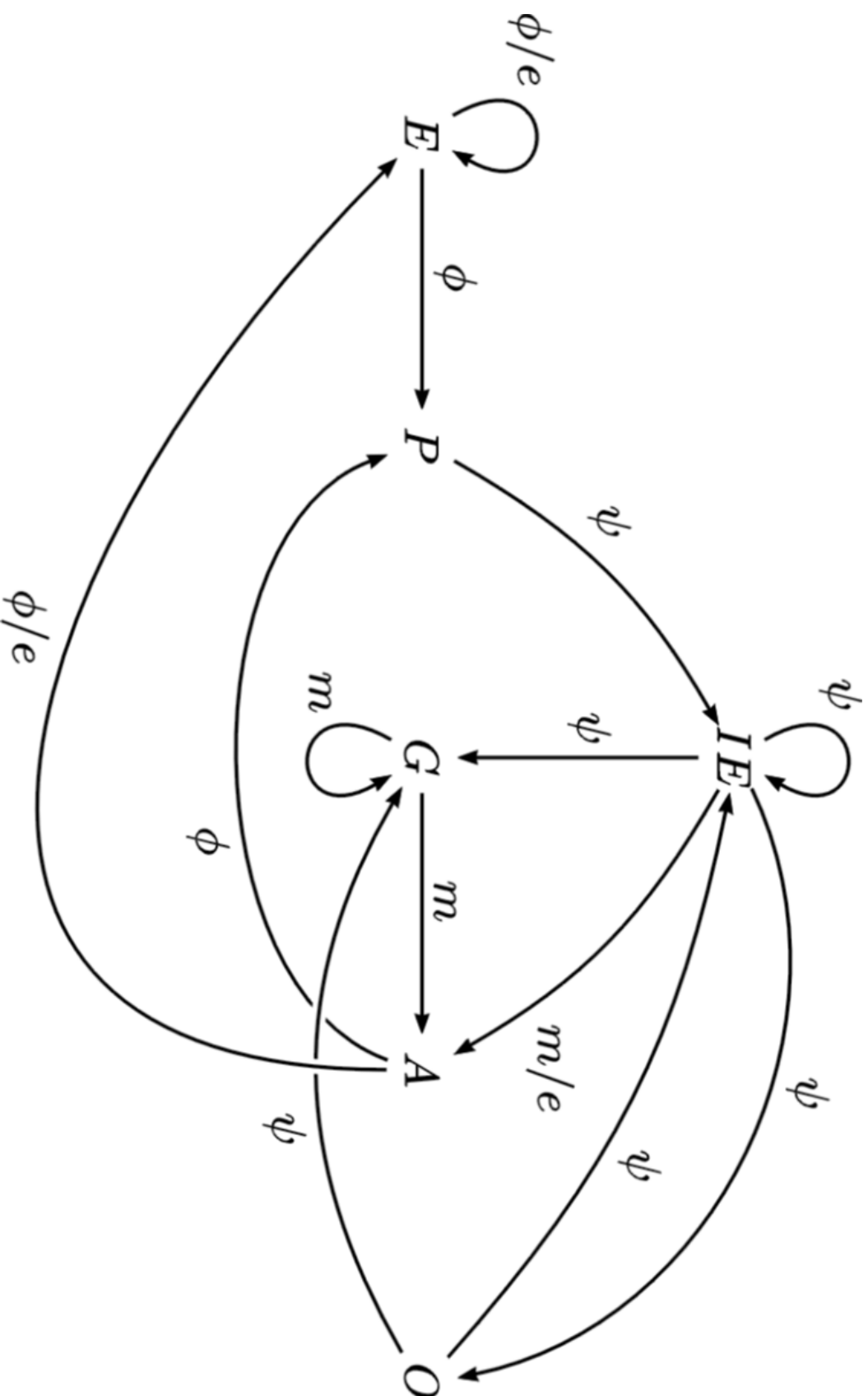
- Story Generation by **Case-Based Reasoning**
 - Pablo Gervais et. al. Universidad Complutense de Madrid, Spain
GEIST: with **Russian Folktales**, 2002
- **Plot-based StoryTeller**:
Mariet Theune , University of Twente, Netherland 2002
Virtual storyteller
- Character-Based Interactive Storytelling
Marc Cavazza et. al., Teesside University, U.K.
 - **Sitcom: HTN plan-based** “Friends” 2002
 - **Merchant of Venice**: Story Generation from **Character’s View** 2010
- Emergent Narrative **based on Social planning**:
Chang and Soo, NTHU, Taiwan 2009
 - Shakespeare’s Othello

PRINCIPLE OF SOCIAL CAUSALITY IN FABULA

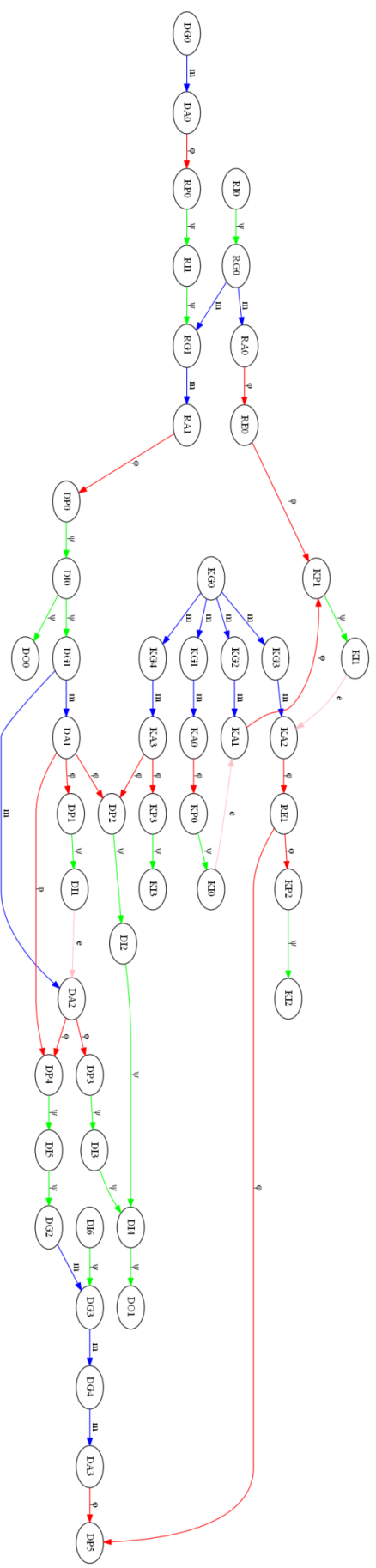
- Causal relations among:
Event, Perception, Action, Goal, Outcome
(states), Internal Events (Mental States)

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FABULA MODEL



STORY PLOT GENERATED ACCORDING TO FABULA



Generating a story from a Fabula

– Virtual Storyteller

16/5/30

- “Princess Lovely”, a story written by Katri Oinonen.
 - Princess Lovely, who lives alone with her father in a big castle.
 - She wants to play with some children she sees outside, but her father, King Mikura, does not allow it.
 - Instead he gives her a little lamb to play with.
 - Princess Lovely loves the lamb very much, but one day it runs away.
 - Lovely searches for the lamb, only to discover that the children she wanted to play with have slaughtered the lamb because they were hungry.

Issues on agent-based story telling systems

- Automatic plot development by characters as intelligent agents
- Control of the character agents to achieve a well-structured plot
- Turning a plot into a narrative using natural language generation
- Story presentation by embodied, speaking agents in a virtual environment
- Involving the user in the story creation process (interactivity)

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Script-based vs Character-based plot generation

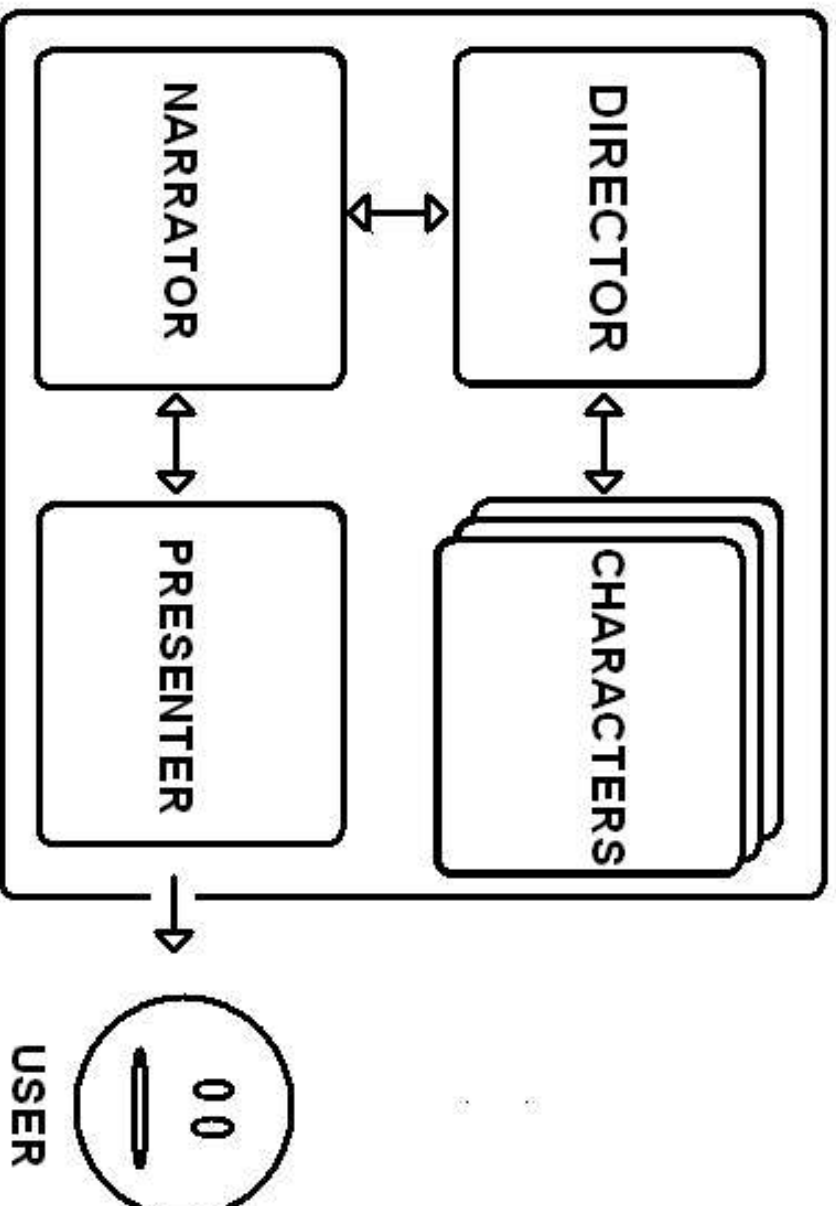
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- How to generate a consistent and well-structured story plot
 - Script-based: Plot is designed by a human authors or an automatic generator based on a story grammars (e.g. Prodd 31 *narratemes* of folk tales)
 - Good: ensure the story in well-structured
 - Bad: consistency in plausibility is low, plot variation is limited
- Character-based: Plot is generated by interactions in autonomous agent characters
 - Good: better chance of well-structured
 - Bad: adequate emergent plot cannot be guaranteed

Intermediate approach to plot generation

- In the Virtual Storyteller, plots are not pre-defined but created by the actions of the characters, guided by a virtual director
- The characters can make plans to achieve their personal goals using story-world knowledge:
 - knowledge about their virtual environment and the actions they can take in it
- A narrator maps a plot into a textual narrative in natural language sentences

Agent architecture of a virtual story teller system



Plot generation— creation as combined efforts

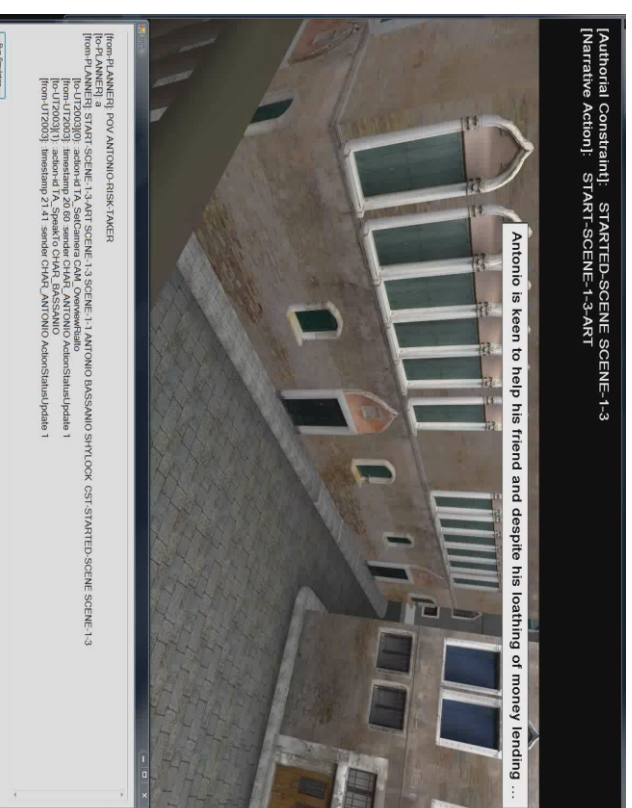
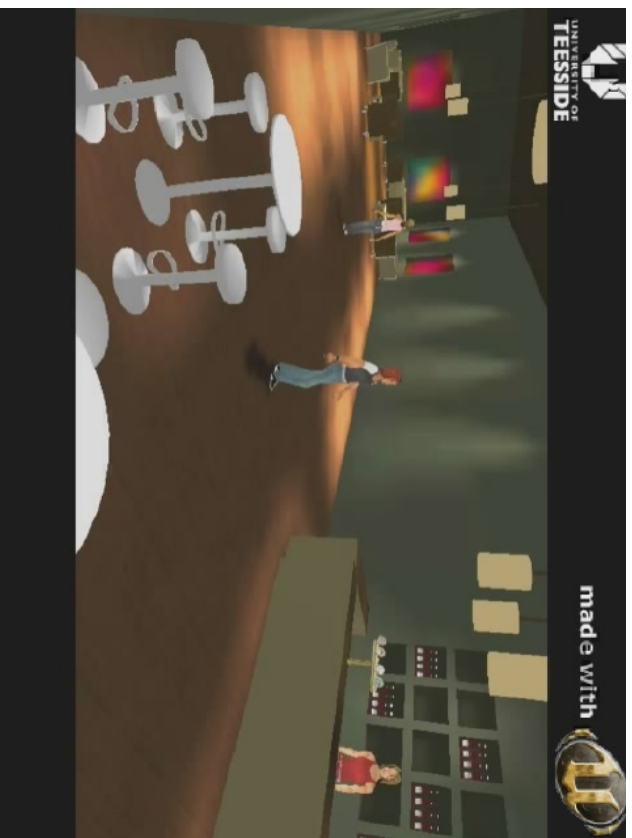
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- The characters themselves ensure the consistency of the plot.
- Director control character's actions via techniques of
 - Environmental: introducing new characters and objects into the story world.
 - Motivational: giving a character a goal to pursue that satisfies the plot.
 - Proscriptive: disallowing a character's intended action.

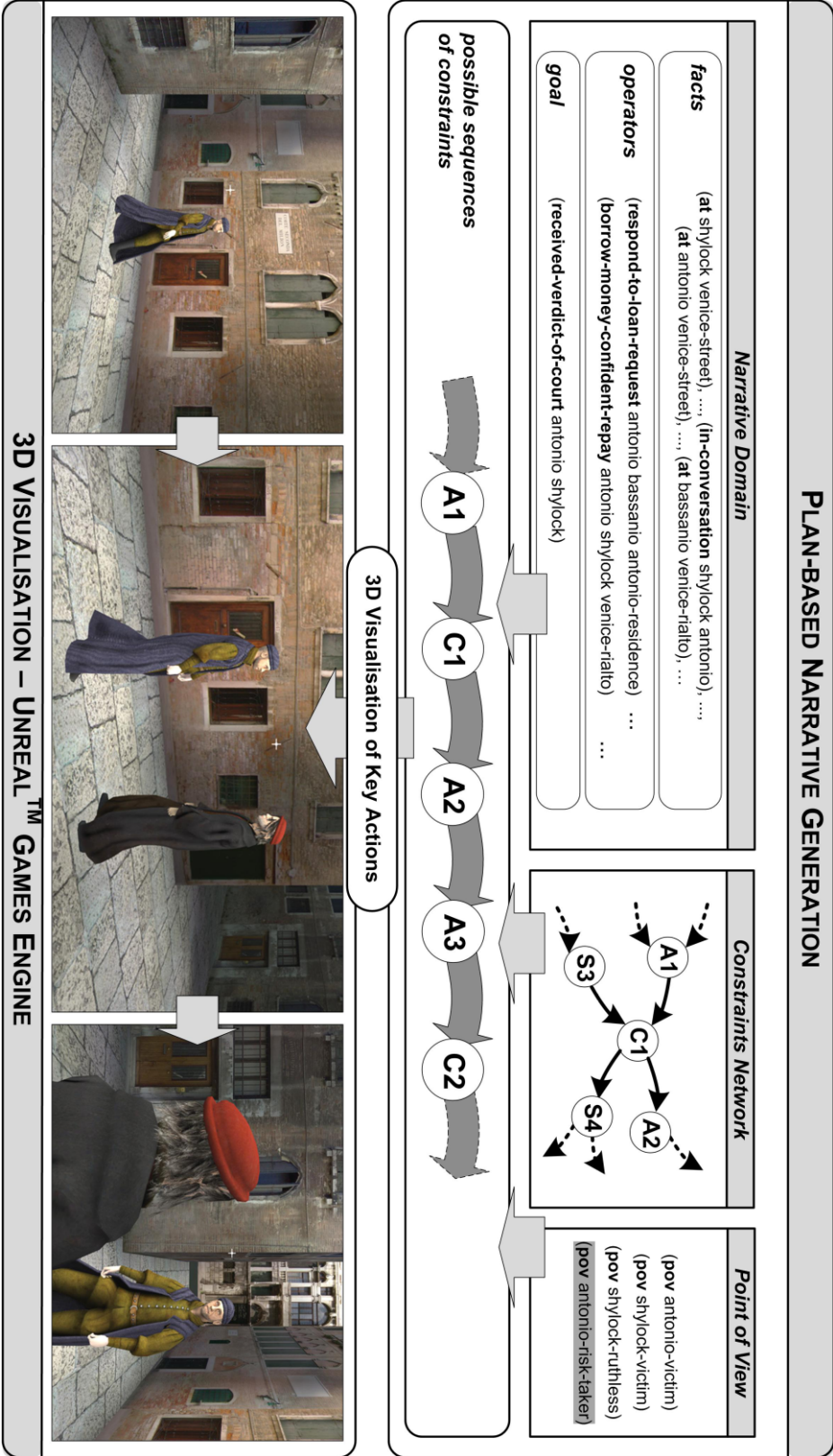
Character-based story generation Cavazza et. al.

- SitCom's Friends 2002
- Shakespeare's drama: Merchant of Venice 2010

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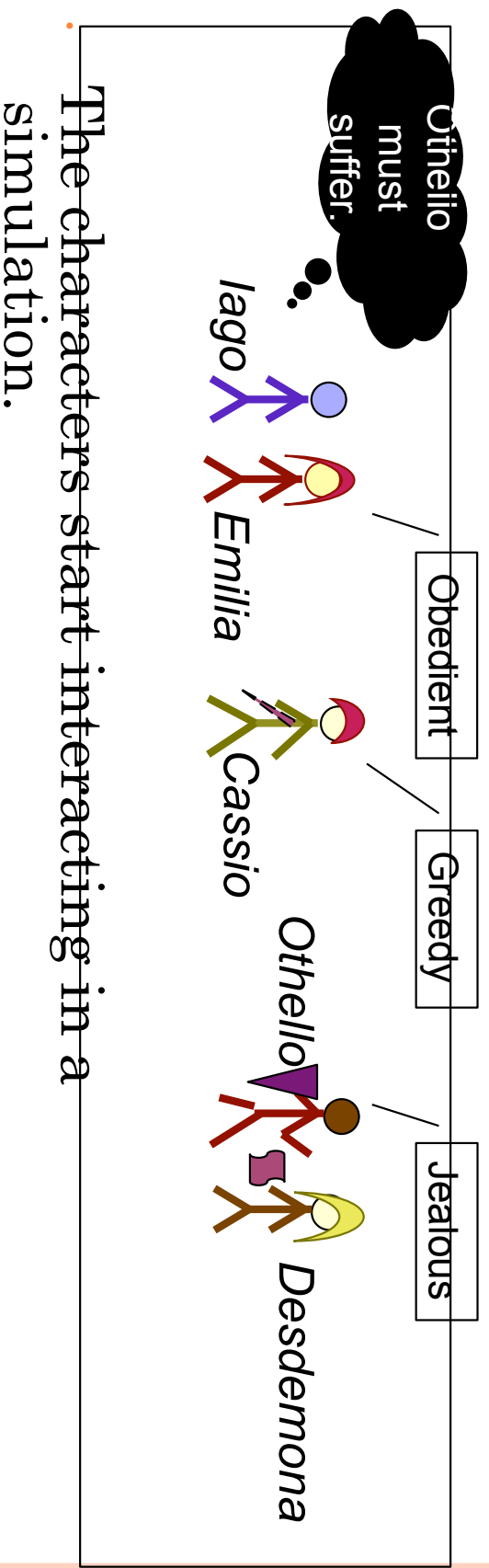


Constraint-based narrative generation



Story Generation: A social-planning based approach

- The author specifies:
 - Initial physical states.
 - Initial mental states of character agents.
 - Including an initial goal of a main character.
 - Model personality of character agents in terms of logic rules.



- The characters start interacting in a simulation.
- The system generates:
 - An total sequence of events that has a story-like quality.



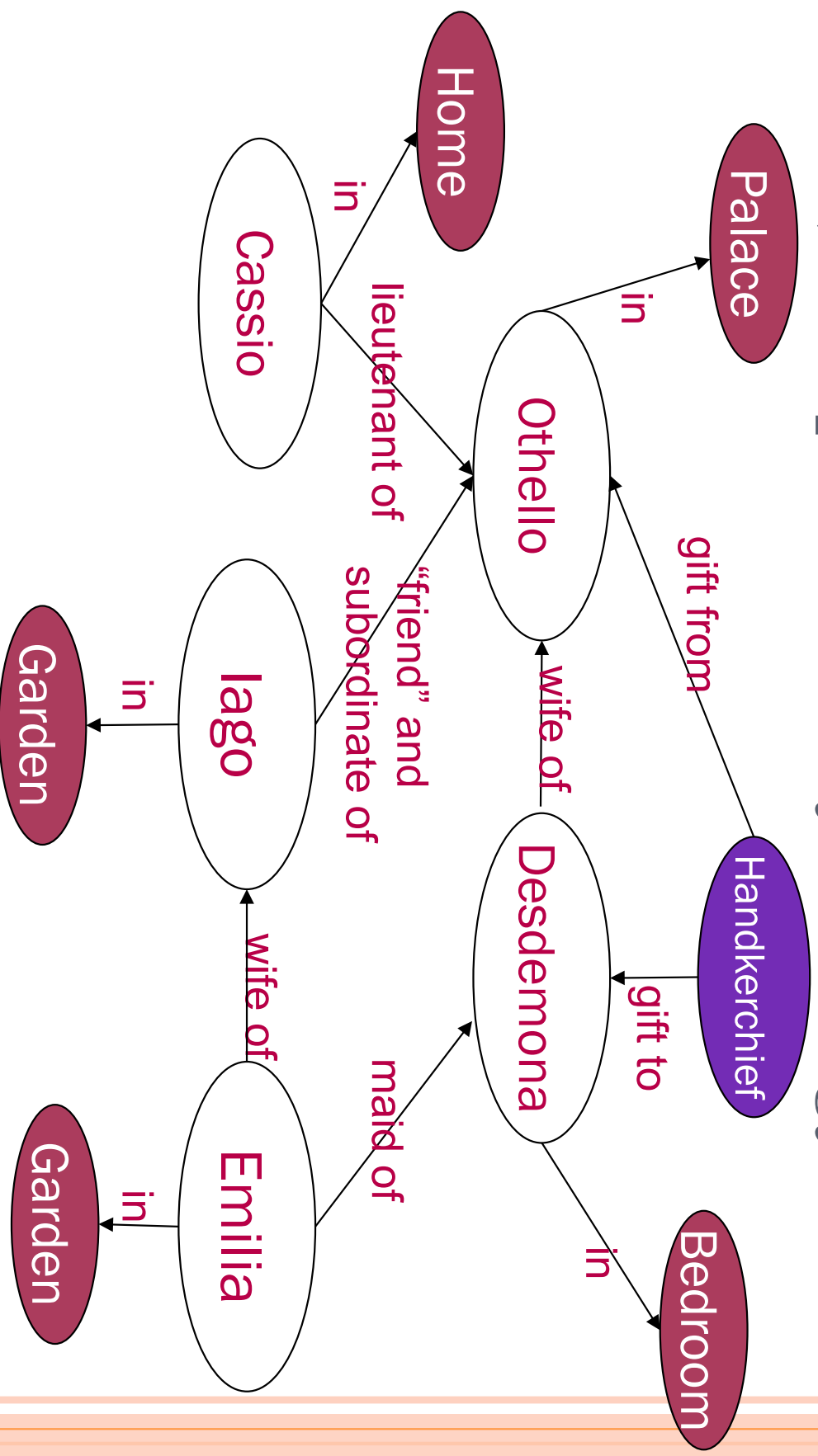
Concepts of Social Planning

- Notion of Social planning:
Social planning is the automated planning process that uses theories of mind to infer the social consequences of an action.

Modeling actions and search thru feasible action space involve multiple agents:

- Heuristic Search Planner (HSP) [Bonet and Geffner, 2001]
- PDDL, the Planning Domain Definition Language [McDermott, 1998].

OTHELLO, simplified story ontology



Cassio: Greedy;

Emilia: Obedient

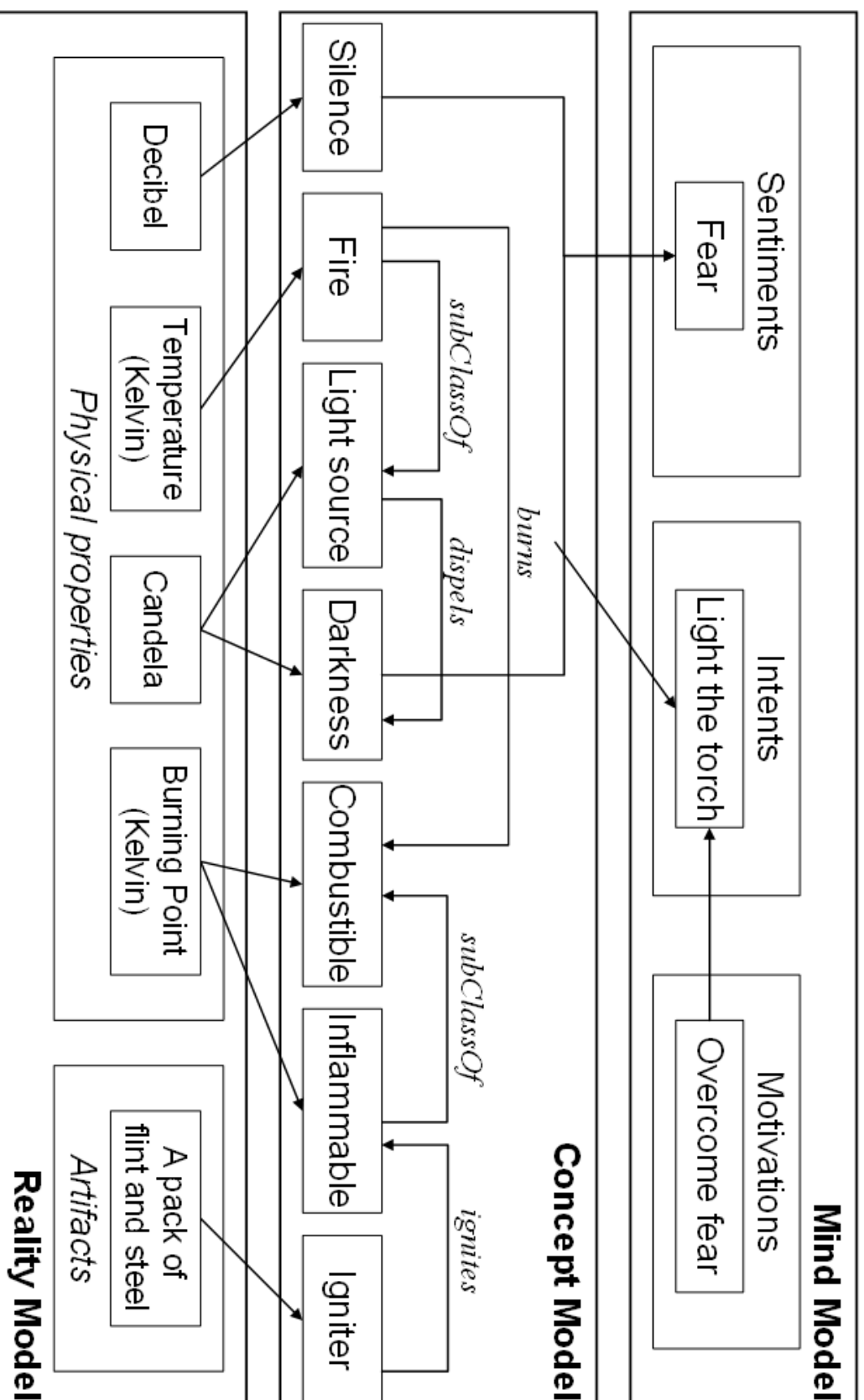
Iago is the main character

Othello: Curious, Jealous

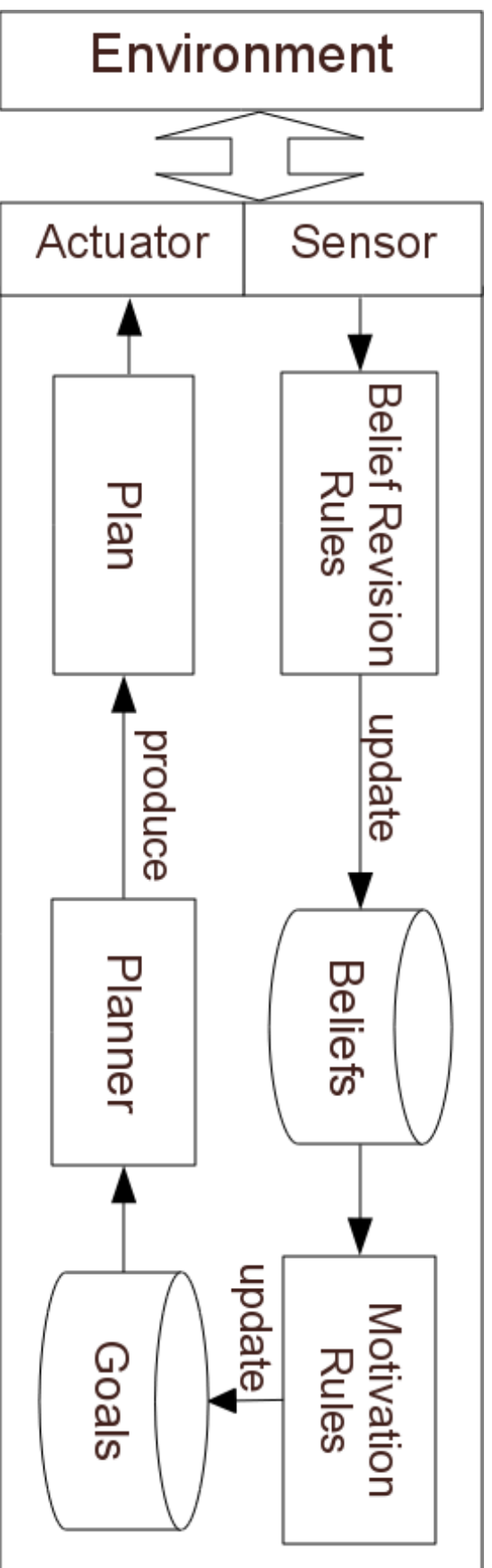
All: Suspicious



Ontology-based virtual environment



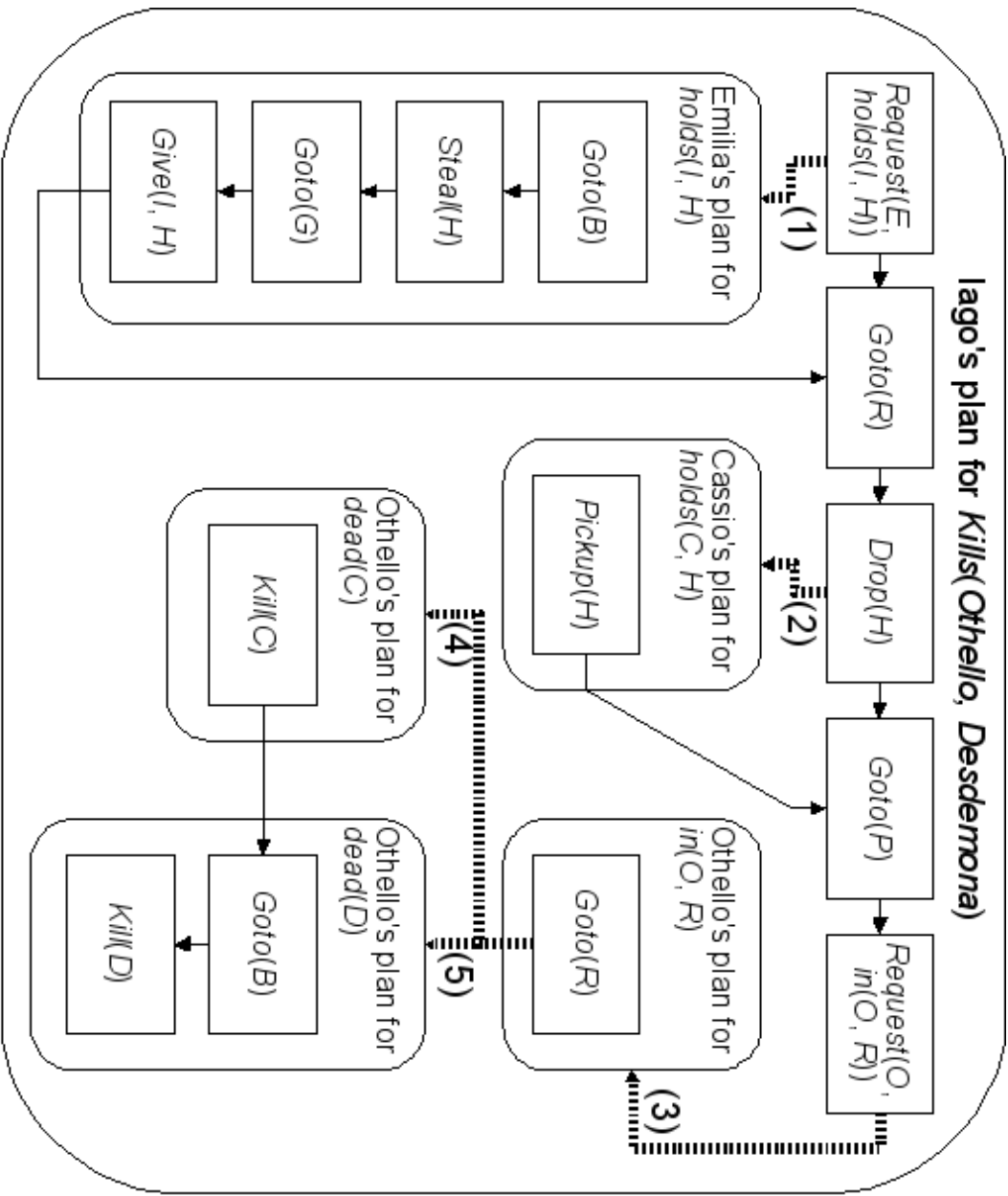
The architecture of I-merge Agents



MENTAL MODEL RULES

Motivation Rules (Generates Goals)	
Personal Motive	Greed : $self(?s) \wedge at(?s, ?l) \wedge at(?o, ?l) \wedge precious(?o) \rightarrow holds(?s, ?o)$ - One will pick up precious things at his or her place.
	Curiosity : $self(?s) \wedge befriends(?s, ?a) \wedge request(?a, ?s, at(?s, ?l)) \rightarrow at(?s, ?l)$ - One will accept a friend's request to visit some place.
	Pursuit : One will give precious things to his or her lover.
Emotion	Wrath : One will desire the death of the killer of his or her lover.
	Jealousy : One will desire the death of both his or her spouse and the lover of the spouse when there is one.
Social Disposition	Obedience : One will accept his or her lover's request to obtain something.
	Belief Revision Rules (Generates Beliefs)
Suspicion :	One who sees a man holding a something another man gave a woman as a gift will believe that the woman deserts the gift-giver for the other man.
Vanity :	A woman will love a man who gives her something precious.
Lust :	A man will love a woman who holds (wears) something precious.

Schematic view of simplified Othello story



Characters: I = Iago, E = Emilia, C = Cassio, O = Othello, D = Desdemona.
Places: G = Garden, B = Bedroom, R = Cassio's residence, P = Palace.
Objects: H = Desdemona's handkerchief.

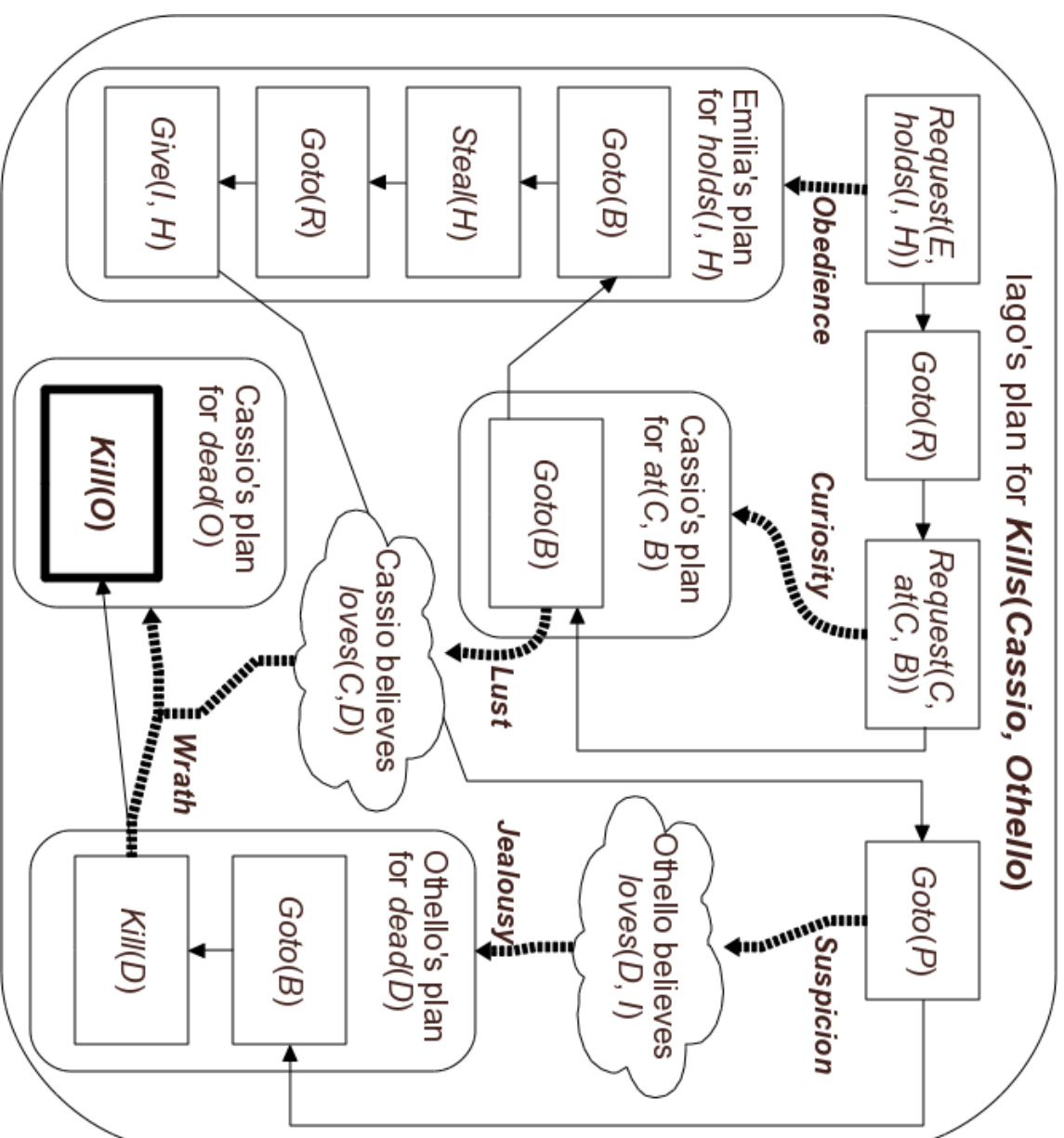
NU#1 from Shakespeare's Othello

Make an agent kill his own wife

16/5/30

- A soldier **Iago** has long resented **General Othello** and plotted to make him suffer by having him kill his own wife, **Desdemona**.
- To carry out his plan, Iago first asked his wife **Emilia** in the garden to steal a handkerchief Othello gave Desdemona as a gift.
- Emilia stole it from Desdemona and gave it to Iago. Iago then deliberately planted the handkerchief in the lieutenant **Cassio's** residence, expecting that Cassio would pick it up, and Cassio did it.
- Othello, under the advice of Iago, then paid a visit to Cassio.
- Seeing the handkerchief on Cassio, he was enraged by jealousy.
- He killed Cassio immediately and ended up going to Desdemona's bedroom to kill her as well.

Social plan representation of NU2.



NU#2: Make other agent to kill an opponent

- Iago resented Othello and plotted to have him killed [The goal of the social plan].
- The villain first asked [Perception leading to Obedience] his obedient wife, Emilia, [Initial state: *hasRule(Emilia, Obedience)* and *isSpouseOf(Emilia, Iago)*] to steal a handkerchief which Othello gave Desdemona as a gift [Initial state: *isGiftTo(Handkerchief, Desdemona)*, *isGiftFrom(Handkerchief, Othello)*]. Iago then went to Cassio's residence to ask [Perception leading to Curiosity] the curious Cassio [Initial state: *hasRule(Cassio, Curiosity)*] to visit Desdemona. Upon seeing her with a handkerchief [Perception leading to Lust], the lustful Cassio [Initial state: *hasRule(Cassio, Lust)*] fell in love with her. Now, Emilia stole the handkerchief for Iago, who then went to the Palace where Othello stayed [Initial state: *at(Othello, Palace)*]. Seeing his lover's handkerchief on Iago [Perception leading to Suspicion], the suspicious Othello [Initial state: *hasRule(Othello, Suspicion)*] believed that Desdemona was now in love with Iago. The jealous Othello [Initial state: *hasRule(Othello, Jealousy)*] thus went to kill Desdemona. Upon witnessing the death of the one he loved [Perception leading to Wrath], the wrathful Cassio [Initial state: *hasRule(Cassio, Wrath)*] revengeed her by killing Othello.

16/5/30

Motivations in social planning

16/5/30

- Changing level of motivation of an virtual character can produce different versions of stories.
- Motivating others is not the only mode of social planning; at least four modes exist:

- **Motivating others.**

Example: making Othello kill Desdemona.

- **De-motivating others** (that is, make a goal disappear).

Example: dissuading Othello from killing Desdemona.

- **Avoiding motivating others.**

Example: Stealing a handkerchief without being seen.

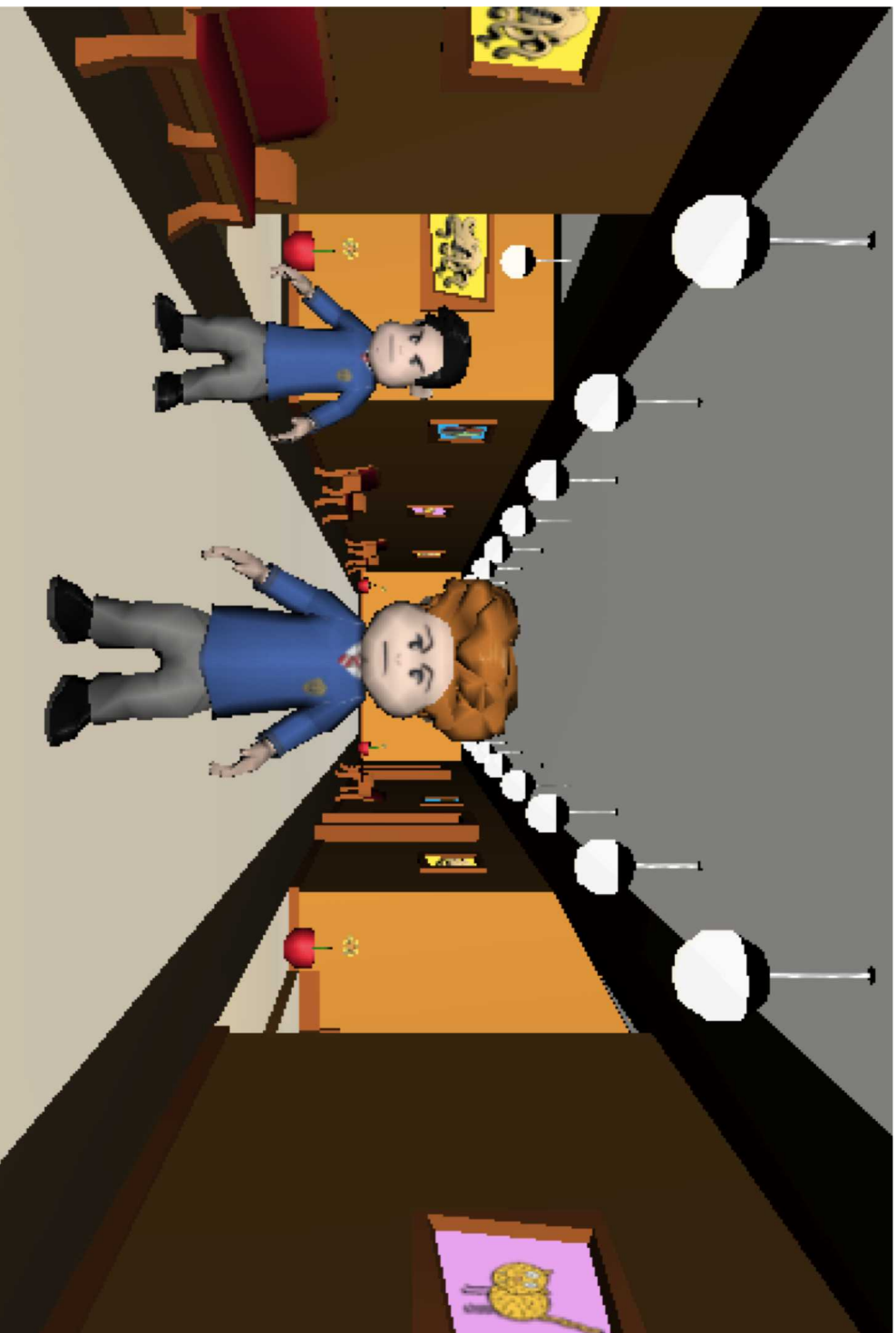
- **Avoiding de-motivating others.**

Example: Making Desdemona love Iago while keep Cassio's interest in her.

Some Agent-based Interactive Dramas -- *with Human involved*

- FearNot! [Aylett and Louchart, 2002-2005]
- Façade [Michael Mateas, 2003-2005]
- Madame Bovary [Cavazza et. Al, 2007]
- Thespian [Mei Si, Stacy Marsella et. al, 2005]

Emergent interactive in FearNot!



- Bully Luke and Victim John
- Emotion responses

FearNot! Project

- Ruth Aylett et. al. at UK and Portugal
- EU FP5 project VICTEC 2002-2005 and extended in the FP6 project eCIRCUS [eCIRCUS 07].
- 3D graphic embedded cartoon-like characters
- Interaction dialogue in terms of text
- FAtiMa agent architecture
- OCC model cognitive appraisal and coping behaviors
- Episode-based and script-based control : believability is under evaluation

Dialogue generation in agent-based social games:

Grace and Trip in *FAÇADE* --- By Michael Mateas 2003



16/5/30

Control discourse in a social game

16/5/30

- A social game in which *players fire off a variety of discourse acts* such as **praise, criticism, flirtation and provocation** in natural language
- The system is divided into affinity game, hot-button game, therapy game.
- The user is supposed to change their (Trip and Grace's) social relations via discourse.
- The system keeps track of the social “tension” level of the story.

FLAUBERT'S NOVEL: MADAME BOVARY -- CAVAZZA



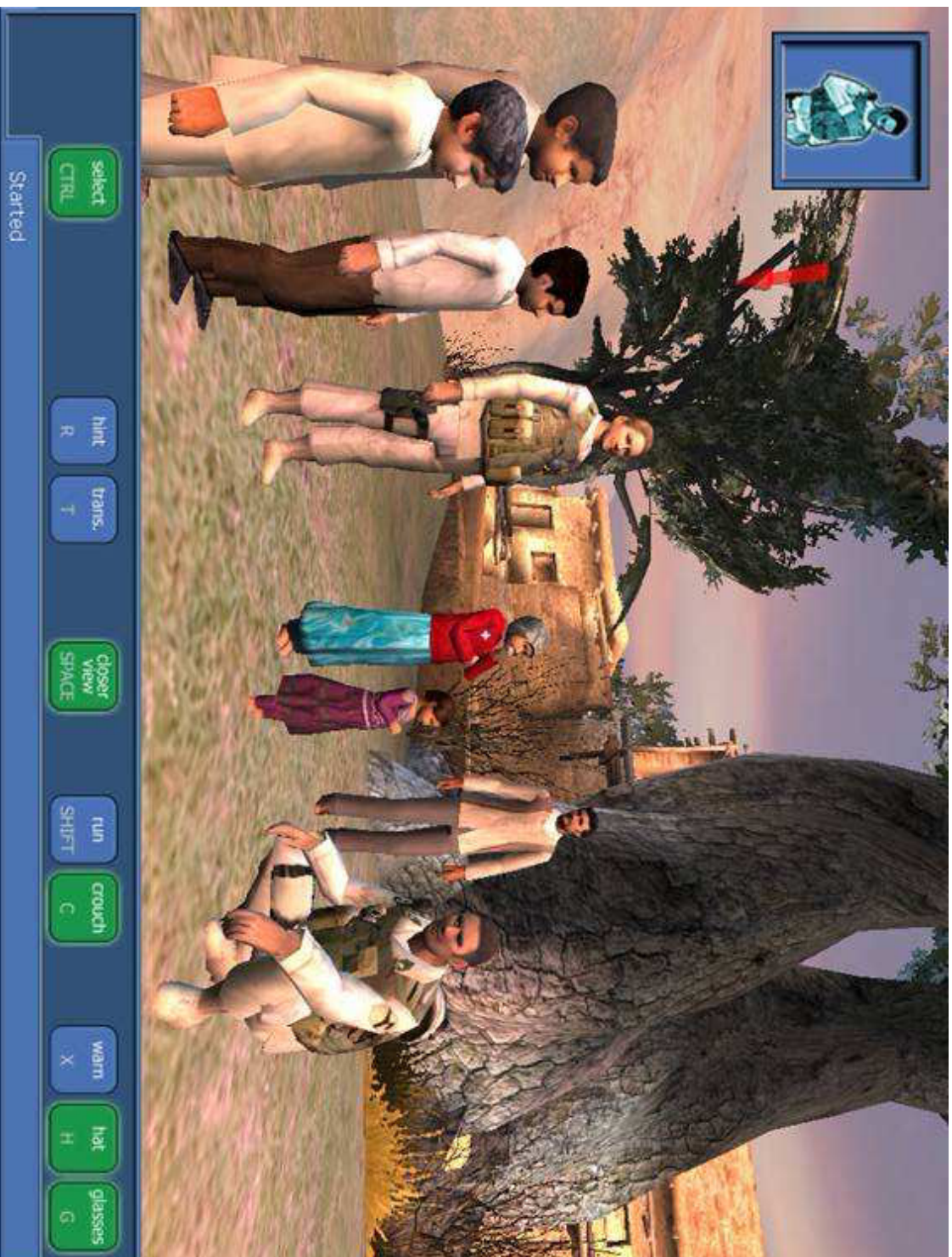
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Interactive Storytelling Engine

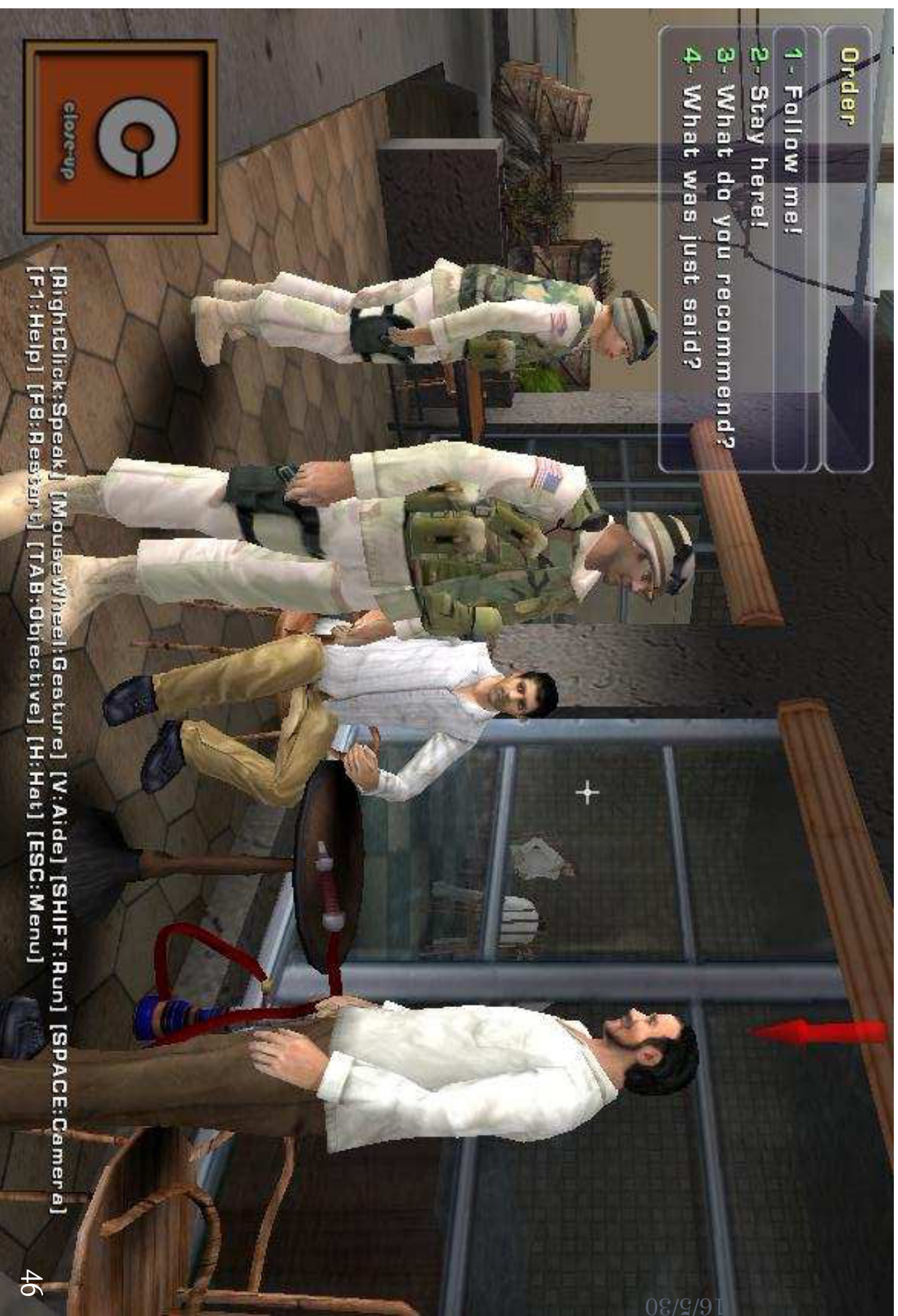
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- 21 feelings each take {low, medium, high}
- Use conjunctions of elementary feelings as goals for the planner
- Allow changes of feelings due to agent actions
- Character-based approach rather than plot-based approach: e.g. character emotion driven
- A HSP (a heuristic search planner)
- Immersive narrative is composed of a succession of real-time animations showing character moving, performing actions, expressions, body attitudes and gestures.

Agent-based Interactive Drama -- **THESPIAN**



16/5/30



Thespian dialogue generation

- Adjacency pair norms
- Turn taking norms: three basic rules on turn-taking behavior
- Conversational flow: begin; body, close
- Affinity control: relation changed over conversation

Pairs and Corresponding Obligations

16/5/30

<u>Speaker 1</u>	<u>Speaker 2</u>	<u>Obligation</u>
Greet	Greet back	Greet back to speaker 1
Bye	Bye	Say "Bye" to speaker 1
Thanks	You are welcome	Say \You are welcome" to speaker 1
Offer X	Accept/Reject X	Either accept or reject X to speaker 1
Request X	Accept/Reject X	Either accept or reject X to speaker 1
Enquiry about X	Inform about X	Inform to speaker 1 about X
Inform information	Acknowledgement	Acknowledgement to speaker 1

Dialogue Generated by THESPIAN Domain

Example

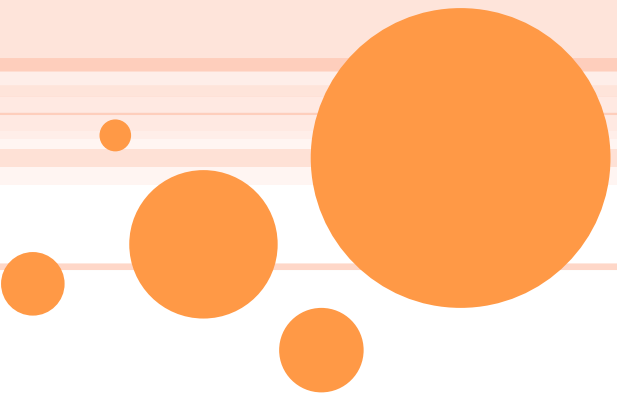
1. Sergeant Smith to Kids: Hello!
2. Xaled to Sergeant Smith: Hello!
3. Hamed to Sergeant Smith: Hello!
4. Kamela to Sergeant Smith: Hello!
5. Sergeant Smith to Xaled: What is your name?
6. Xaled to Sergeant Smith: My name is Xaled.
7. Xaled to Sergeant Smith: What is your name?
8. Sergeant Smith to Xaled: My name is Mike.
9. Sergeant Smith to Xaled: How are you?
10. Xaled to Sergeant Smith: I am ne.
11. Xaled to Sergeant Smith: Are you an American?
12. Sergeant Smith to Xaled: Yes, I am an American.
13. Sergeant Smith to Xaled: I am learning Pashto.

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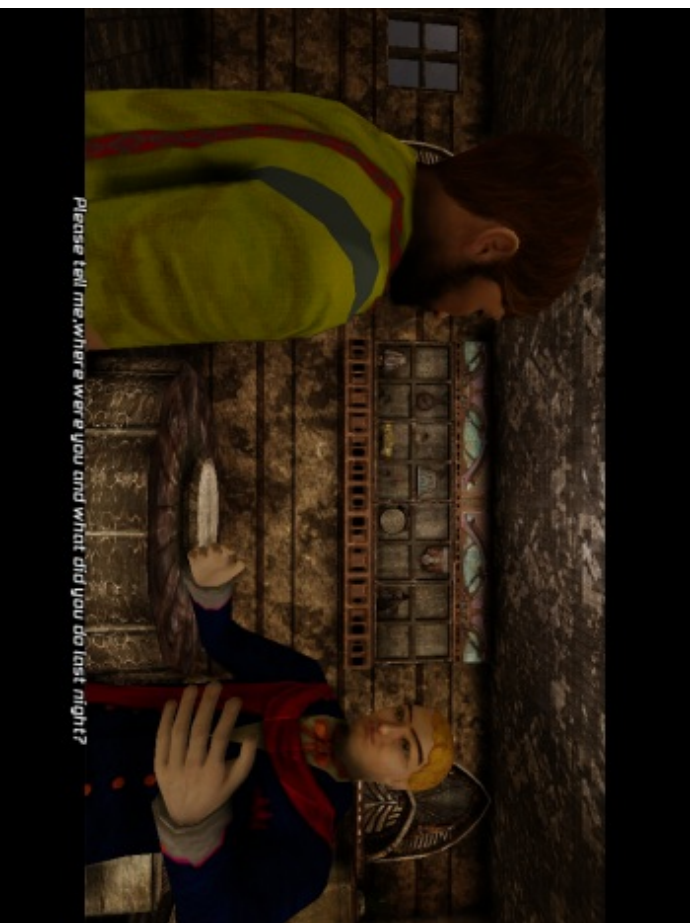
Issues of control of story development

- a definite beginning, middle and ending [Theune, et al, 2003]
- arcs of tension [Mateas and Stern, 2003]
- suspension [Cheong and Young, 2008]
- emotional intensity [Aylett and Louchart, 2008]
- Change motivation sensitivity (personality?) [Chang and Soo 2009]
- etc.

HOW CAN VIRTUAL AGENTS BECOME SOCIAL-AWARE WHILE INTERACTING WITH HUMANS?



SOCIAL-AWARENESS IN INTERACTING VIRTUAL DRAMA



WHAT IS SOCIAL-AWARE?

- Knowing the belief, desire, and intention of other agents in a given social context
- Knowing the emotion state, mood, personality of other agents in a given social context
- Knowing the social relations, roles of agents in a given social context
- Knowing the commitments, social norms /convention or even morality and culture of agents in a human society



AGENT INTERACTIONS AS COMMUNICATION ACTS

- Speech act vs Physical act
 - Mental state changes vs physical environment change
- Verbal interaction vs non-verbal interaction
- To facilitate the efficiency and effectiveness of communication between multi-agents
- Agent communication language design in agent research community assume Grice maxims.



GRICE MAXIMS REVISITED

- Assume agents are cooperative in the conversation (interaction):
 - Quality: truthfully (honest)
 - Quantity: convey as much information as required (precise)
 - Relation: convey relevant information when necessary
 - Manner: appropriate (avoid ambiguity, clear and orderly)



BARRIERS OF SOCIAL-AWARENESS IN HUMAN INTERACTIONS

- In true human communication and interactions, humans cannot always follow Grice maxims due to:
 - Not always be cooperative
 - Not always certain (to find the exact terms to express oneself) and precise (can sometimes lengthy and redundant)
 - Might have erroneous beliefs even with truthful intention/implicature
 - Might not be polite (not follow the conventional protocol or even hostile)
 - Might be malicious or negative pragmatic such as irony or sarcasm
 - Not always be honest sometimes cheating
 - Might not express directly rather use indirect hints (sometimes due to politeness)
 - May have ambiguous or multiple intentions
 - Might not always be consistent between true intention and actual behaviors (intentionally or non-intentionally)



DIFFICULTIES OF SOCIAL AWARENESS WITHOUT SOCIAL CONTEXT MODELS

- We cannot verify human mental states exactly in a directly-measurable manner
- With sensor data alone, we cannot easily distinguish social mental states of agents without context background information and social models
 - Too many ambiguities/exceptions without background contexts
 - Many context assumptions are needed for sensor mining that are not explicit in domain applications.



DETECT EMOTIONS FROM DIALOGUE TEXT (WITHOUT DIALOGUE CONTEXT MODEL)

- Motivations:
 - OCC model provides an appraisal model for emotion reasoning and affection computation
 - The emotion inference is based on the appraisal logic on objects, agents and events toward agent's own preference and goal
 - An event fails may lead to a disappointment emotion
 - But it is hard to apply to dialogue without much domain-specific knowledge
 - It must be modulated by social relations and agent's individual personality



TESTING DIALOGUE SENTENCE CORPUS

- micro blog website “Plurk” (<http://www.plurk.com/>) with tagged emotion icons (we collected the corpus with Plurk unofficial API in 2009

Table 4. Examples of sentences in our corpus with emotion icons.

S/N	Raw sentence, Phrase structures, Thematic roles, Emotion icon and 3-class Emotion Polarity
S1	條紋衫會讓人頭暈 (Striped shirts will make people feel dizzy). [NP [NP 條紋] [ADVP 會] [V 讓] [NP 人] [VP 頭暈], [Ca 條紋衫] [Ma 會] [V 讓] [Be 人] [Po 頭暈]] (striped shirts) [will] [make] [people] [feel dizzy]), , negative
S2	明天氣即將變暖 (The weather will be warm tomorrow), [NP 明天] [NP 天氣] [ADVP 即將] [V 變暖], [Ti 明天] [Tt 天氣] [Ma 即將] [V 變暖]] ([tomorrow] [the weather] [will] [be warm]), , positive
S3	從昨天晚上不舒服到早上 (I do not feel comfortable from last night to this morning). [PP 從 [NP [NP 昨天] 晚上]] [ADVP 不] [V 舒服] [PP 到 [NP 早上]], [P 從] [So 昨天晚上] [Ma 不] [V 舒服] [P 到] [Go 早上]] ([from] [last night] [do not] [feel comfortable] [to] [this morning]), , negative
S4	半夜一群貓在外面大合唱 (A crowd of cats sung in choruses in the garden at midnight). [NP 半夜] [NP [NEU 一]] [NF 群] 貓 [PP 在 [NP 外面]] [ADVP 大] [V 合唱], [Ti 半夜] [Qu 一群貓] [P 在] [Lo 外面] [Ma 大] [V 合唱]] ([at midnight] [a crowd of cats] [in] [the garden] [in chorus] [sing]), , negative

Table 5. The rule of transforming emotion icons into positive/negative/neutral polarities.

[illegible]

DICTIONARIES AND THESAURUS

Table 1. The taxonomy in Tongyicil in thesaurus

Code	Subject	Syntactic role
A	Person	Noun
B	Object	Noun
Ca	Time	Noun
Cb	Place	Noun
D	Abstraction	Noun
E	Feature	Adjective
F	Motion	Verb
G	Thought	Verb
H	Action	Verb
I	State	Verb
J	Relationship/Phenomenon	Transitive Verb
K	Auxiliary Words	Function Word
L	Honorary Words	Noun

Table 2. Examples of words in HOPT dictionary.

Word	Class	Polarity
我 (I, me, etc)	h (human)	+
對手 (opponent)	h (human)	-
作者 (author)	h (human)	0
興趣 (interest)	o (object)	+
暴行 (brutalities)	o (object)	-
藥物 (medicine)	o (object)	0
天堂 (heaven)	p (place)	+
地獄 (hell)	p (place)	-
廁所 (toilet)	p (place)	0
明天 (tomorrow)	t (time)	+
昨天 (yesterday)	t (time)	-
今天 (today)	t (time)	0

Table 3. Examples of words in the verb dictionary.

Word	Pos	Polarity
住院 (be hospitalized)	VA	-
搖 (shack)	VAC	0
求婚 (make an offer of marriage)	VB	+
處罰 (punish)	VC	-
檢查 (check)	VE	0
停止 (stop)	VF	-
高興 (happy)	VH	+
困擾 (perplex)	VHC	-
心儀 (admire in the heart)	VI	+
忘記 (forget)	VK	-



LEARNING THEMATIC ROLES FIRST MIGHT

HELP EMOTION CLASSIFICATION

Table 8. Thematic roles are useful in detecting emotions.

Thematic role	Examples
Time	I am scared <i>now</i> . (fear) I was scared <i>at that time</i> . (relief)
Location	I ate my dinner <i>at home</i> . (neutral) I ate my dinner <i>in jail</i> . (negative)
Theme	I got <i>a slap</i> from my friend. (negative) I got <i>a gift</i> from my friend. (positive)
Beneficiary	I cleaned the room <i>for my girl friend</i> . (positive) I cleaned the room <i>for a corrupt official</i> . (negative)
Instrument	I opened the door <i>with my key</i> . (neutral) I opened the door <i>with the stolen key</i> . (negative)
Source	I come back to my normal life <i>from hell</i> . (positive) I come back to my normal life <i>from heaven</i> . (negative)



ATTRIBUTES IN LEARNING EMOTION CLASSIFICATION

- 53 attributes in total to train a neural network classifier

Table 6. The features used and their rationales (similar attributes has the same designation).

Attribute	Rationales and examples
HasNeuFront	To identify Qu, e.g. the thematic role of “[NP [NEU 六] [NF 小時]]” is Qu.
HasNg	To identify Ti, Go, and So, e.g. the thematic role of “[NP 今天 [NGS 起]]” is So.
pBefore, NpBehind, PreNpCode*, PostNpCode*	To increase the performance (we supply the semantic features of three noun phrases at a time).
Passive, SubObj	To identify Ag, Th, etc. For example, in “我(I) 被 (by) 老師 (the teacher) 處罰 (punish)” (I was punished by the teacher.), the thematic roles of “我” and “老師” are Th and Ag respectively.
InPrep, In**Prep	To disambiguate the thematic roles of a noun phrase in a PP.
Animate	To identify Ag, e.g. “學生” (student), “猴子” (monkey).
Code*, VCode*	To provide the semantic features of the current parsing noun phrase and the verb of the sentence based on the semantic coding in the Chinese thesaurus.
VSense	To identify Ex, e.g. “看” (see), “希望” (wish), “得知” (know).
VSocial, VMove, VTurn	To identify Go, e.g. “恭喜” (congratulate), “到達” (arrive at), “變成” (turn into).

ADABOOST + MULTI-LAYER PERCEPTRON LEARNING

- AdaBoost introduced by Freund and Schapire in 1995 is the first proposed boosting algorithm. The name “Ada”, the abbreviation of “adaptive” means AdaBoost can adapt to the error rates of all weak classifiers.
- We use feedforward 3-layer neural networks as weak classifiers for learning thematic roles and emotions.
- From the input side of a neural network classifier, we need to enter semantic features of lexicon in the sentences, while the output we collect the predicted thematic role class or emotion type.



THEMATIC ROLE LEARNING WITH 3-LAYER PERCEPTRONS

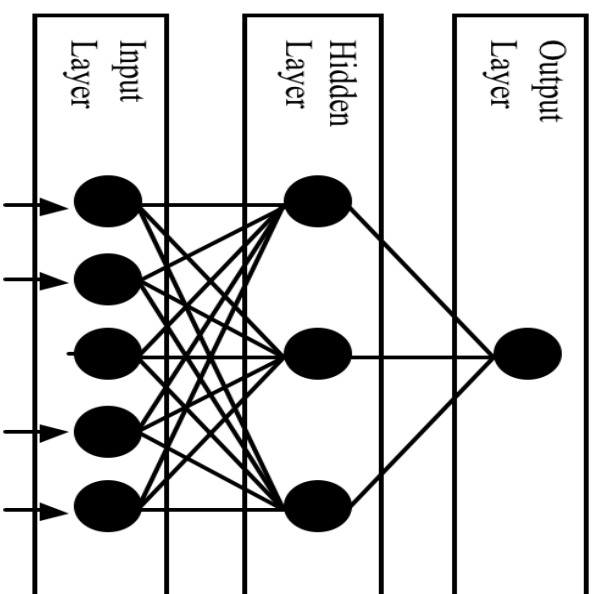


Figure 2.A 3-layer MLP neural network architecture



ADABOOST LEARNING

Input

Sequence of training examples

$$L = \{(x_1, y_1), \dots, (x_N, y_N), x \in X, y \in Y\}$$

The number of boosting rounds T

Weak classifier $H : X \rightarrow Y$

Distribution D over training examples

for $i=1, \dots, N$

$$D_1(i) : 1/N$$

for $t=1, \dots, T$

Sample with distribution D_t from L to get L_t

Train H_t with L_t

$$\varepsilon = \sum_{i=1}^N [D_t(i) \cdot \|H_t(x_i) \neq y_i\|]$$

If $\varepsilon \geq 0.5$, then set $T = T - 1$ and abort loop

$$\alpha_t = \ln((1 - \varepsilon) / \varepsilon)$$

for $i=1, \dots, N$

$$D_{t+1}(i) = D_t(i) \cdot e^{-\alpha_t \cdot \|H_t(x_i) = y_i\|} / Z_t$$

where Z_t is a normalization factor makes D_{t+1} be a distribution

Output

$$\arg \max_{y \in Y} \sum_{t=1}^T [\alpha_t \cdot \|H_t(x) = y\|]$$

$y \in Y$

Note: For any proposition P , if P is true, $\|P\| = 1$, otherwise $\|P\| = 0$.



PERFORMANCE:

THEMATIC ROLE LEARNING

Table 11. Average precisions and recalls of classification of thematic roles in percentage in the preliminary evaluation via the 10-fold cross-validation.

Precision													
Ag	Th	Be	Go	So	Ti	Lo	In	Ca	Re	Qu	Ex	St	
75	89	78	71	89	95	70	93	71	58	96	79	78	
Recall													
Ag	Th	Be	Go	So	Ti	Lo	In	Ca	Re	Qu	Ex	St	
77	91	66	58	53	97	65	78	67	61	95	77	79	

Table 12. Average precisions and recalls of classification of thematic roles in percentage via the 10-fold cross-validation after adding some features associated with verbs.

Precision													
Ag	Th	Be	Go	So	Ti	Lo	In	Ca	Re	Qu	Ex	St	
76	90	81	78	100	95	78	87	85	94	97	83	86	
Recall													
Ag	Th	Be	Go	So	Ti	Lo	In	Ca	Re	Qu	Ex	St	
82	92	69	69	73	98	65	72	73	89	91	85	80	



3-CLASSES EMOTION CLASSIFICATION



The number of sentence examples for 3-class emotion detection

Table 14. Average precisions and recalls of classification of 3-class emotions in percentage via the 10-fold cross-validation using AdaBoost.M1 + NeuralNetworks.

Precision		
Positive	Negative	Neutral
80	84	33
Recall		
Positive	Negative	Neutral
83	87	17



MULTI-EMOTION CLASSIFICATION

Table 15. The number of examples in each emotion class.

Emotion icon	# of examples
	170
	131
	104
	122
	67
	51

Table 16. Average precisions and recalls of classification in percentage on multi-class emotions via the 10-fold cross-validation.

Precision					
					
51	42	40	35	28	13
Recall					
					
52	44	51	33	24	8



PERFORMANCE EVALUATION

Table 17. The inter-categories that are easiest to be misclassified each other.

Emotion icon	Misclassified as
	
	
	
	
	

Table 18. Average accuracies of emotion classifiers with different weak learners in 3-class emotion detection (positive, negative, and neutral) over five experiments via the 10-fold cross-validation.

Algorithm	Accuracy
AdaBoost.M1 + MultilayerPerceptrons (withthematic roles)	80.7147 %
AdaBoost.M1 + MultilayerPerceptrons (without thematic roles)	77.4919 %

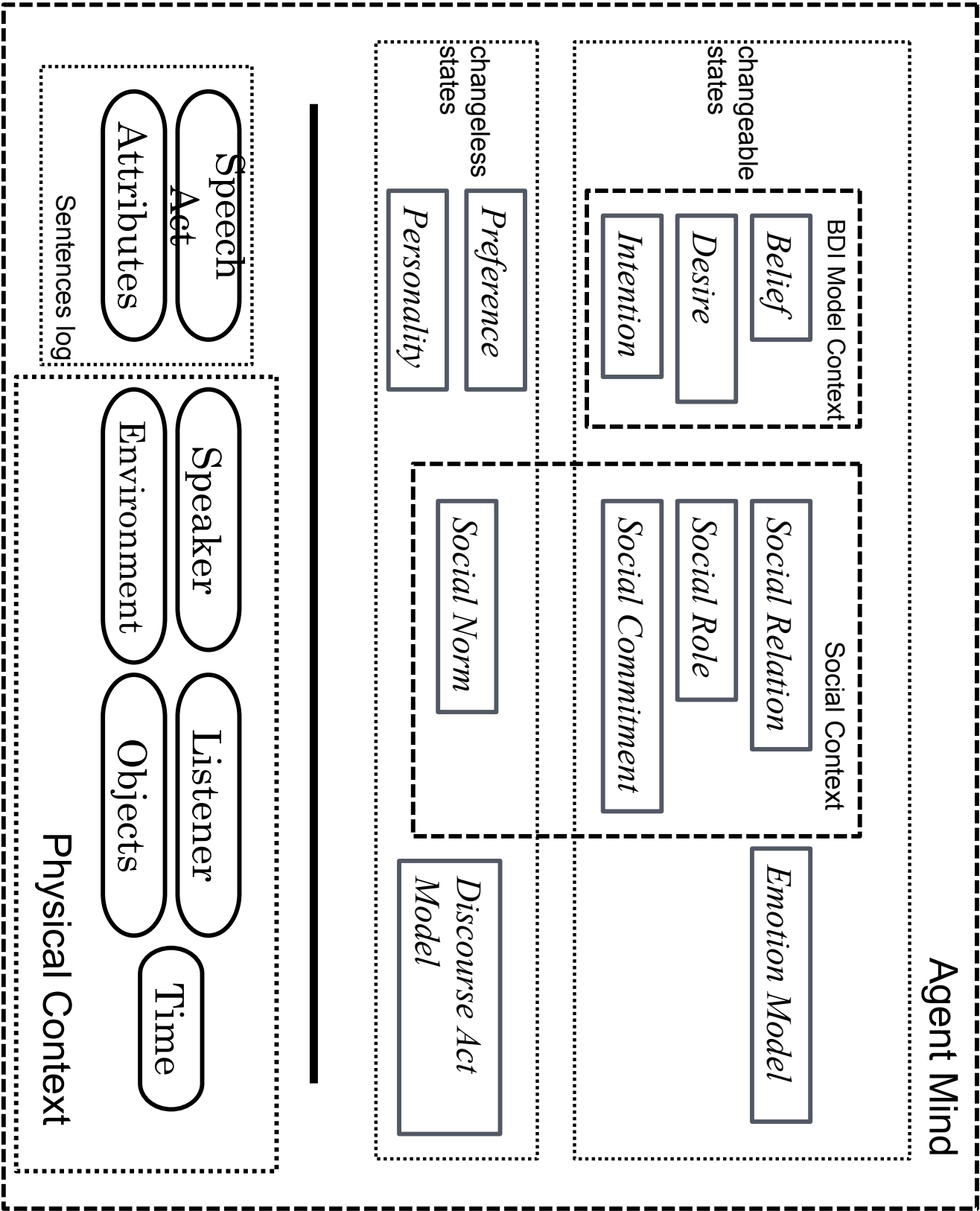


A case Study-- DETECTION OF EMOTION AND INTENTION FROM A SEQUENCE OF SPEECH ACTS WITH SOCIAL CONTEXTS

- We collected a dialogue sequence from a script in a famous movie Doubt which is a 2008 film adapted from John Patrick Shanley's Pulitzer Prize winning fictional stage play *Doubt: A Parable*.
- In a Bronx Catholic school in 1964 a popular priest's ambiguous relationship with a troubled 12 year old black student is questioned by the school's principal.
- In the movie there is a scene where the characters Father Flynn, Sister Aloysius and Sister James were having an argument.

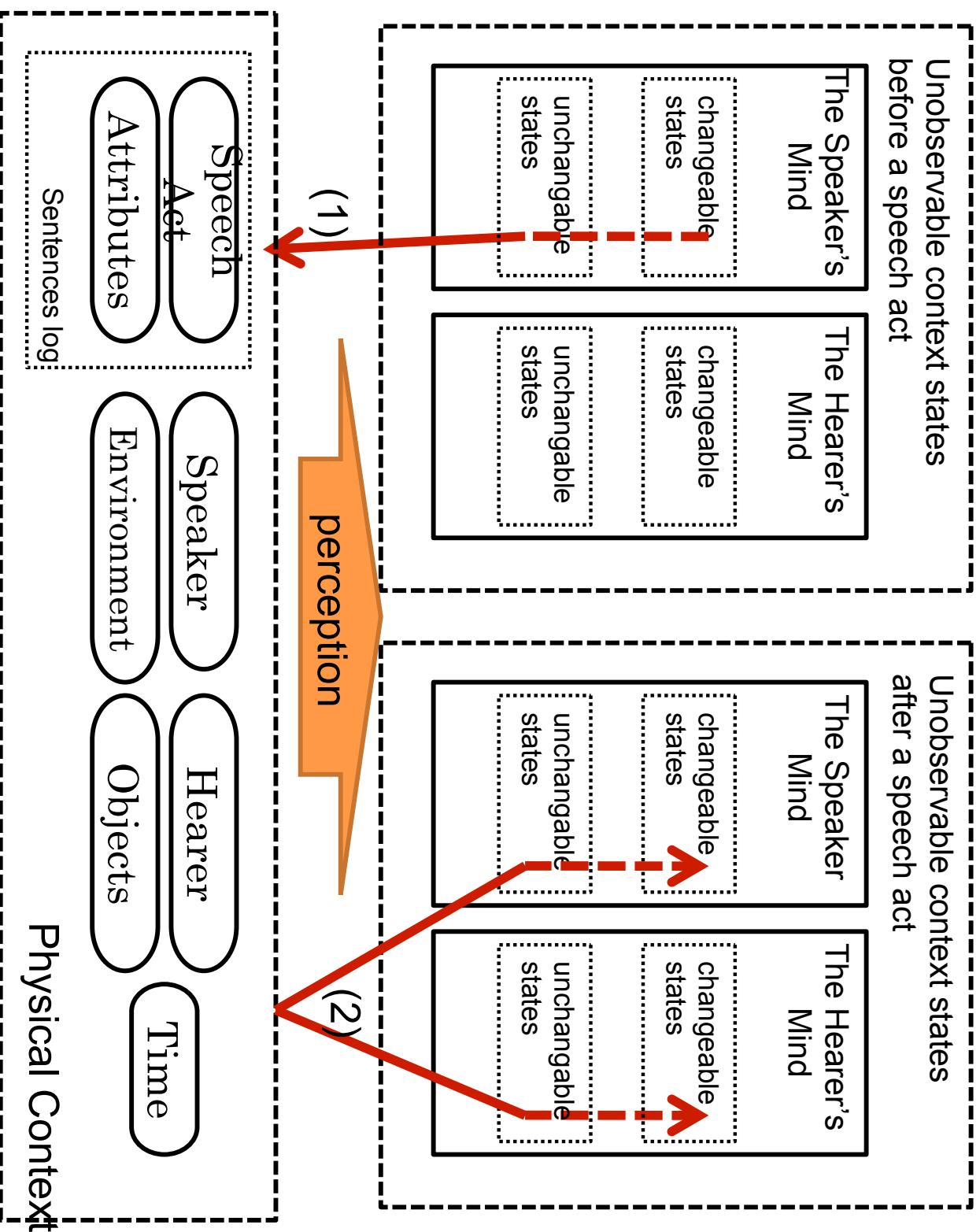


DIALOGUE CONTEXTS



MODEL A SPEECH ACT AS SOCIAL CONTEXT

CHANGE



THE SPEECH ACT MODEL

The speech act of a dialogue sentence can be treated as an action.

- The speech act affects the mental states of speakers and listeners
- A speech act consists of *triggering conditions* and *effects*.

Speaker *orders* a listener to do something



Speaker has a *hope* emotion



Speaker has a *fear* emotion



The listener has a social commitment – *To do something*

Speaker has a social role with the listener: *be_a_chief_to*

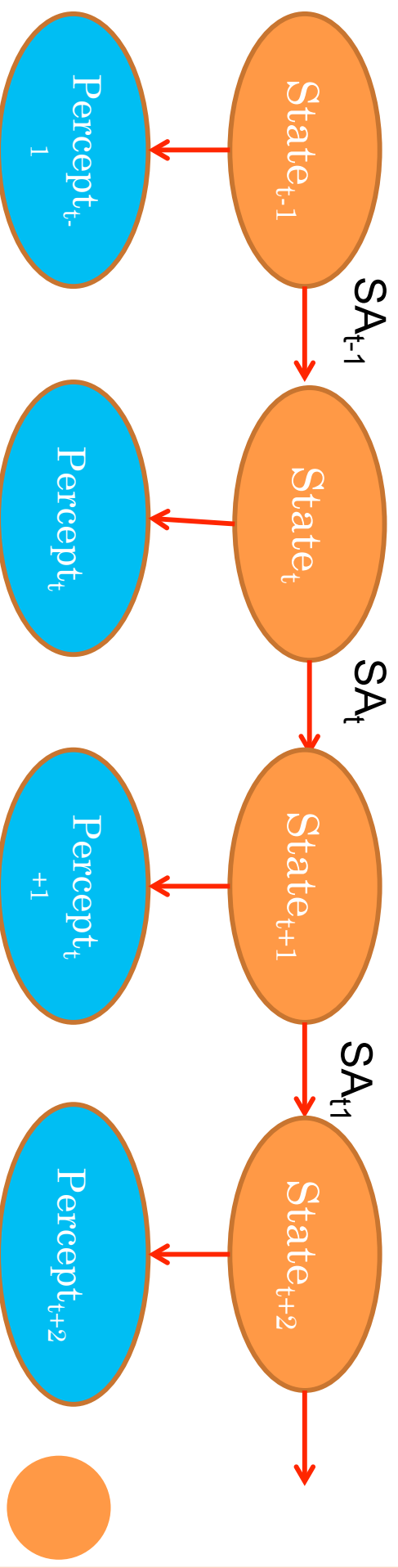


Uses *order*

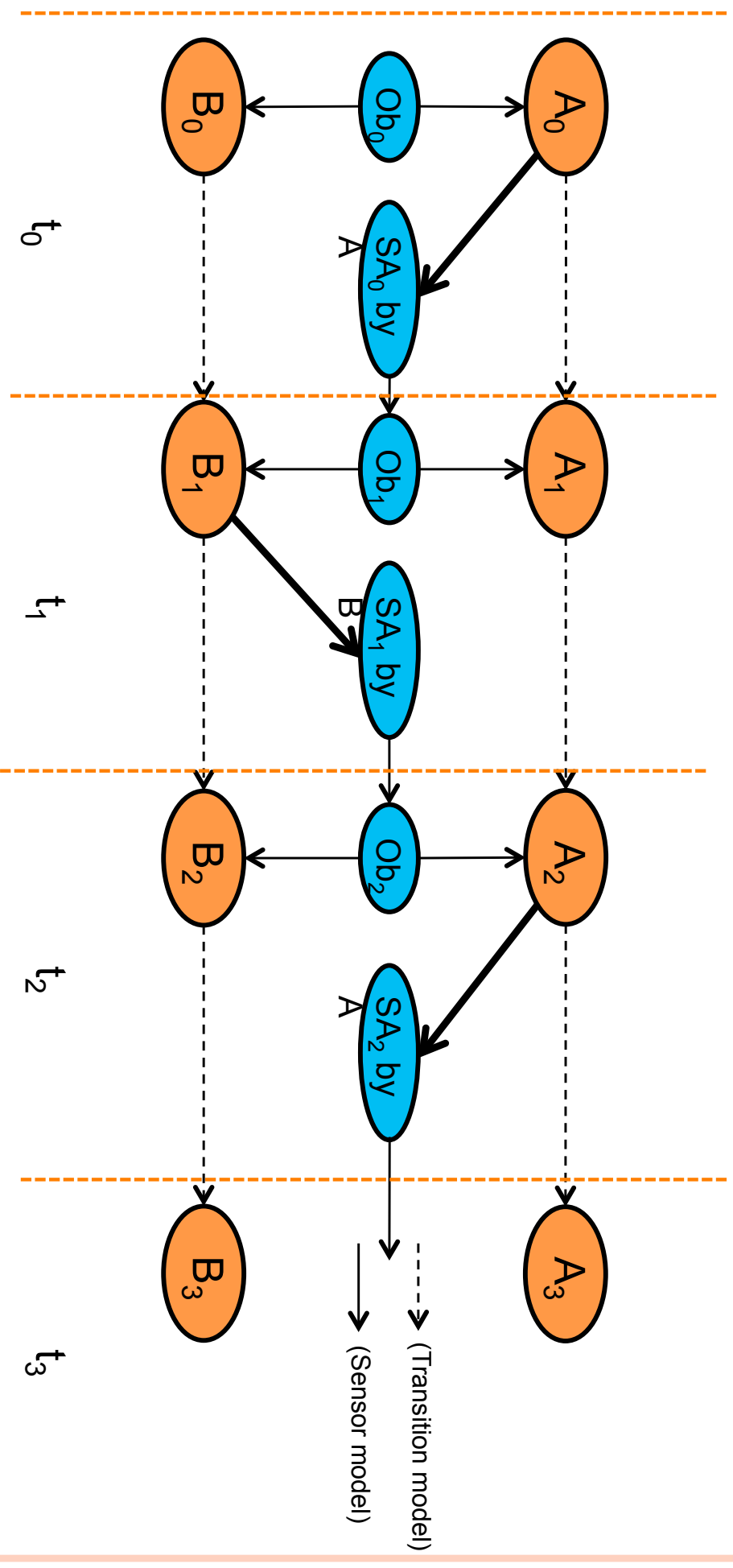


DYNAMIC BAYESIAN NETWORKS TO REPRESENT DIALOGUE ACTS

- Dynamic Bayesian network with time slices.
 - To represent the temporal dependency.
 - Repeated transition model and sensor model:
 $P(E_t | X_t)$ and $P(X_{t+1} | X_t)$
- Directed graphical model of stochastic processes.
 - Prediction, Rollup and Estimation.*



DBN MODEL IN TWO AGENTS' DIALOGUE



- A_n and B_n are dialogue contexts states in time slice n of the agents A and B .
- Any physical states in the environment are assumed to be observable by all agents.

DBN CONDITIONAL PROBABILITIES IN 2-AGENT DIALOGUE

- The observable data (Ob_t) that includes the speech act (SA_t) is the sensor model that reflects states of observable dialogue contexts evidence E_t , and the speech act could also affects the dialogue context.

$$P(E_t | X_t)$$

- The effects of speech act could cause the dialogue context to change with probability $P(X_t | X_{t-1}, SA_{t-1})$ in which SA_t is speech act at dialogue step t .



SEMANTIC VS PRAGMATIC SPEECH ACTS

- We assume:

All agents use the semantic speech acts only in the dialogue.

- A semantic speech act :
The basic type of speech act used by the speaker.
- A pragmatic speech act :
The listener interprets the semantic speech act using current dialogue contexts.

Semantic speech acts	Corresponding pragmatic speech acts
Ask	Ask, Interrogate, Interpellate, Threaten
Request	Request, Beg, Order, Threaten,, Propose,
Inform	Inform, Lie.
Reply	Reply, Reject, Accept, Controvert, Not-intimidated



MODEL A SPEECH ACT WITH PROBABILITY (1)

- A *request* by a boss could be an *order*.
- A boss's *request* could be an *order*.
- Every rule has exceptions.

with the speech act: *request*

Speaker has a social
role with the listener:

be_a_boss_to

Use an *order*

logic model

0.8

Use an *order*

0.02

Use a *beg*

probability model

Speaker has a social
role with the listener:

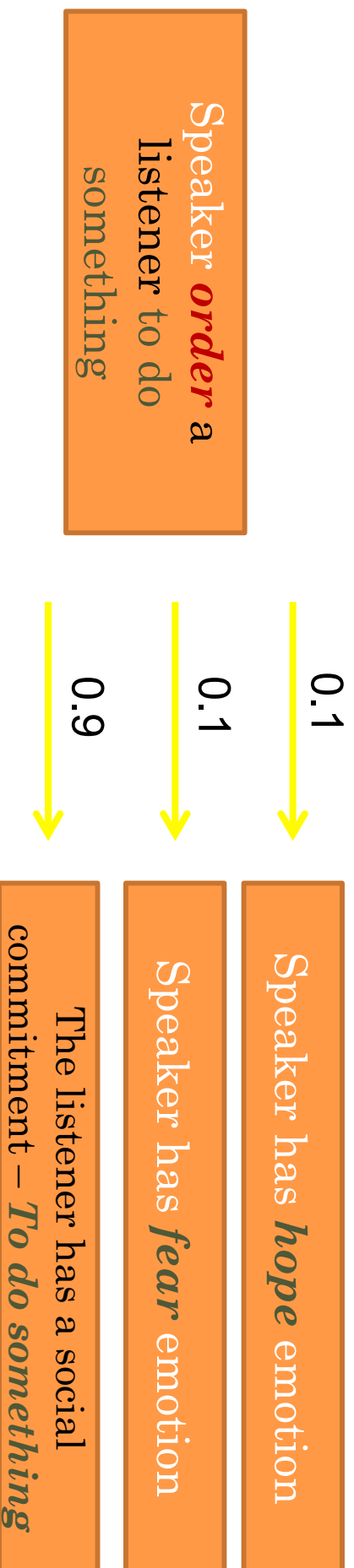
be_a_boss_to

0.18

Use a *request, threaten, propose*

MODEL A SPEECH ACT WITH PROBABILITY -- EFFECTS

- Effects of a speech act can be also attached with probabilities.
- The effects of an *order* :
 - The speaker has emotion *hope* and *fear*.
 - The listener has a social commitment – *To do something*.



MODEL A SPEECH ACT WITH PROBABILITY -- EFFECT

- The different pragmatic speech acts have different effects.

Speaker *order* a listener to do something

10%

Speaker has *hope* emotion

10%

Speaker has *fear* emotion

90%

The listener has a social commitment – *To do something*

Speaker *beg* a listener to do something

90%

Speaker has *hope* emotion

90%

Speaker has *fear* emotion

10%

The listener has a social commitment – *To do something*

Model of speech acts

- We modeled 21 speech acts occurred in the conversation in the Doubt script of 53 dialogue sentences: *Announce, Controvert, Dissatisfied, Recount, Ask, Reply, Interrogate, Reject, Censure, Rebut, Say-goodbye, Accuse, Agree-with, Request, Propose, Be-glad, Threaten; Not-intimidated, Pride, Praise, and Grumble.*
- We implemented DBN to support uncertain reasoning forward and backward from various contexts via dialogue.



IMPLEMENTED DIALOGUE CONTEXTS

- Dialogue Contexts
 - 7 negative emotions and one positive emotion.
 - *Shame, Remorse, Fear-confirmed, Reproach, Disappointment Anger, Fear and Hope*
 - 2 social relations
 - *Friendly_with, Hostile_with*
 - 2 social roles
 - *Be_authority_to, Be_subordinate_to*
 - 2 discourse act models
 - *Ask_Reply, Request_Reply,*
 - 2 personalities
 - *Hide_emotions, Be_nice_to_everyone*
 - Several physical contexts
 - *Speak_to, Listen_from, Time_slice, Location(nearby, far, can_not_hear)*



CORPUS OF 53 DIALOGUE SENTENCES

#1 Aloysius: “The boy’s well-being is my responsibility.”

#2 Flynn: “His well-being is not a issue.”

#3 Aloysius: “I’m not satisfied that that is true.”

...

#50 Aloysius: “Intolerance”

#51 Flynn: “That’s right. I’m not pleased with how you handled this.”

#52 Flynn: “Sister”

#53 Flynn: “Sister”



PRAGMATIC SPEECH ACT CLASSIFICATIONS IN 53 SENTENCES IN DOUBT

Semantic speech acts	Corresponding pragmatic speech acts
<i>Ask(e)</i>	<i>Ask(e), Interrogate(g), Interpellate(v), Threaten(q)</i>
<i>Request(n)</i>	<i>Request(n), Order(w), Propose(o),</i>
<i>Reply(f)</i>	<i>Reply(f), Reject(h), Accept(x), Controvert(b), Threaten(q), Not-intimidated(r)</i>
<i>Recount(d)</i>	<i>Recount(d), Censure(i), Accuse(l)</i>
<i>Announce(a)</i>	<i>Announce(a)</i>
<i>Dissatisfied(c)</i>	<i>Dissatisfied(c)</i>
<i>Rebut(j)</i>	<i>Rebut(j)</i>
<i>Agree_with(m)</i>	<i>Agree_with(m)</i>
<i>Say_goodbye(k)</i>	<i>Say_goodbye(k)</i>
<i>Be_glad(p)</i>	<i>Be_glad(p)</i>
<i>Pride(x)</i>	<i>Pride(s)</i>
<i>Praise(t)</i>	<i>Praise(t)</i>
<i>Grumble(u)</i>	<i>Grumble(u)</i>

PRAGMATIC VS SEMANTIC SPEECH ACTS

Request

- Request
- Order
- Propose

Recount

- Recount
- Censure
- Accuse

Ask

- Ask
- Interrogate
- Interpellate
- Threaten

Reply

- Reply
- Reject
- Accept
- Controvert
- Threaten
- Not-intimidated



AN EXAMPLE MODEL OF A PRAGMATIC SPEECH ACT

○ *A threaten(ask) B*

$P(\text{threaten} \mid A_emotion_anger) = 0.5$

$P(\text{threaten} \mid A_emotion_hope) = 0.5$

$P(\text{threaten} \mid A_relation_hostile_with_B) = 0.35$

$P(A_relation_hostile_with_B \mid \text{threaten}) = 0.35$

$P(B_relation_hostile_with_A \mid \text{threaten}) = 0.65$

$P(B_emotion_fear \mid \text{threaten}) = 0.5$

$P(B_emotion_anger \mid \text{threaten}) = 0.65$

$P(B_emotion_anger \mid \text{threaten}) = 0.65$

$P(B_will_reply \mid \text{threaten}) = 0.85$

Triggering
conditions

Effects



DEFAULT STATES OF DIALOGUE CONTEXTS

- There are some background knowledge in the scenario.
 - About personality
 - Flynn, Aloysius and James : Hide_emotions
 - James : Be_nice_to_everyone
 - Default location
 - Flynn *nearby* Aloysius
 - Flynn *nearby* James
 - Aloysius *nearby* James
 - Discourse act model



SPEECH ACTS INPUT SEQUENCES

A. Semantic speech acts input sequence:

(21a)(12b)(21c)(21d)(13e)(31f)(21e)(12e)(21f)(12f)(21e)
(12f)(21e)(32e)(23f)(12i)(21j)(12e)(21f)(21e)(12e)(21f)
(21e)(12h)(12k)(13k)(21d)(12e)(31f)(21e)(12n)(21h)
(31n)(12n)(21h)(12f)(31p)(21e)(12f)(12f)(31t)(13s)(31t)
(32e)(23f)(12m)(12e)(21f)(12d)(21u)(12l)(12k)(12k)

B. Pragmatic speech acts input sequence:

(21a)(12b)(21c)(21b)(13e)(31f)(21g)(12e)(21f)(12h)(21g)
(12h)(21g)(32e)(23f)(12i)(21j)(12e)(21f)(21g)(12e)(21f)
(21g)(12h)(12k)(13k)(21l)(12e)(31m)(21g)(12n)(21h)
(31n)(12n)(21h)(12b)(31p)(21q)(12r)(12q)(31t)(13s)(31t)
(32e)(23f)(12m)(12e)(21f)(12d)(21u)(12l)(12k)(12k)

○ (Speaker, Listener, A speech act) – *physical contexts omitted*

- 1 = Father Flynn, 2 = Sister Aloysius, 3 = Sister James



SPEECH ACTS USED IN 53 DIALOGUE SENTENCES IN DOUBT

a: Announce,	b: Controvert,	c: Dissatisfied,	d: Recount,
e: Ask,	f: Reply,	g: Interrogate,	h: Reject,
i: Censure,	j: Rebut,	k: Say-goodbye,	l: Accuse,
m: Agree-with,	n: Request,	o: Propose,	p: Be-glad,
q: Threaten;	r: Not-intimidated,	s: Pride,	t: Praise,
u: Grumble.			



EXPERIMENT 1 :

PRAGMATIC DETECTION WITH DIALOGUE CONTEXTS

- We split the dialogue into two parts (26 sentences and 27 sentences).
- Train with the first half (26 sentences with pragmatic speech acts) to learn partially the dialogue contexts states.
- Test with the second half dialogue with semantic speech act and ...
 - With the information of dialogue contexts from the first half dialogue.
 - With only default information of dialogue contexts.

Only 19 of 27 sentences in the second half dialogue have multiple possible pragmatic speech acts in the classification.

Correct data		Accuracy
Results with context info.	lemgnfnnhbpbqrqtstefmefdu kk	15/19
Results w/o context info.	demenfnnf p ef t stefmefdu dkk	11/19

EXPERIMENT 2-1 :

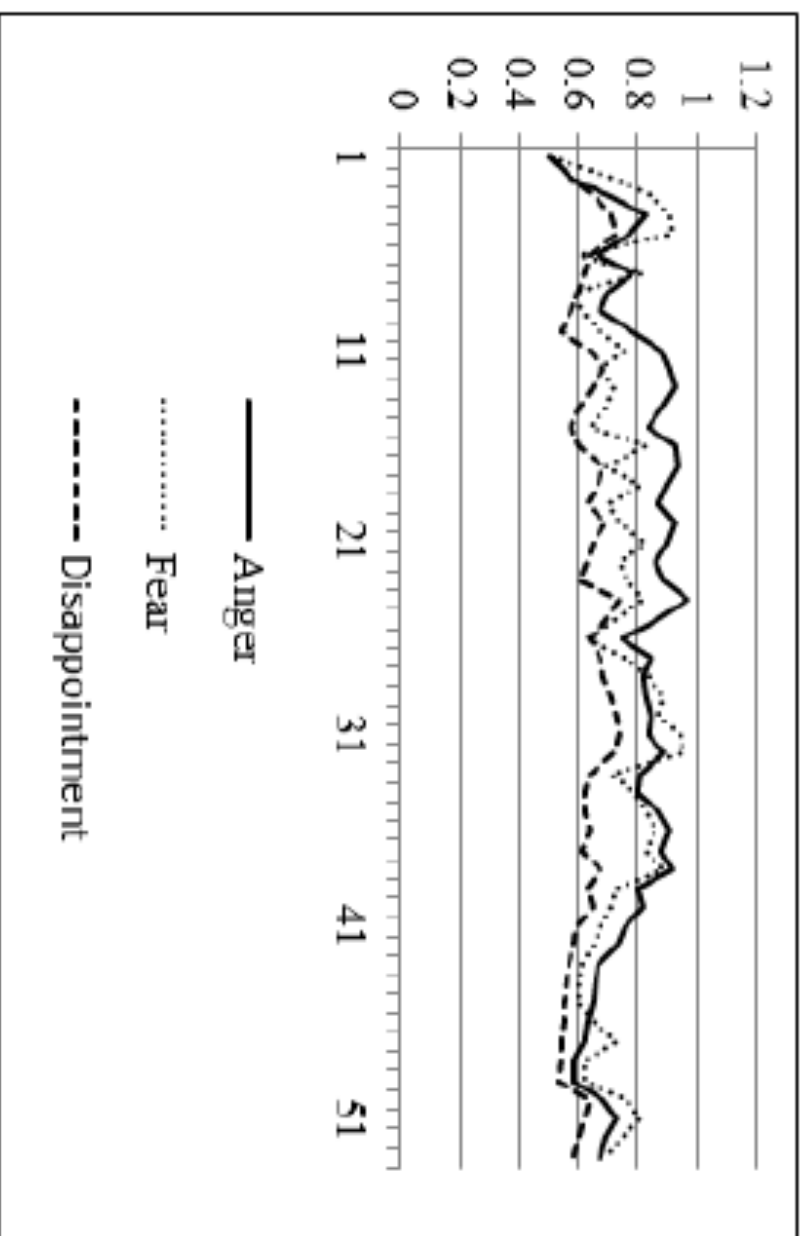
THE MOST LIKELY PRAGMATIC SPEECH ACTS SEQUENCE.

- We tested the whole dialogue with semantic speech acts and default information of dialogue contexts.
 - The result of the second half dialogue has two more wrong detections than experiment 1.
- We used semantic speech acts in this experiment, and used pragmatic speech acts in experiment 1.

Only 40 of 53 sentences in the dialogue have multiple possible pragmatic speech acts in the classification.

Correct data	abcbefgefhgngfijefgetghkk lemgnfnhbpqrqtstefmedulkk	Probability Accuracy
Result	adcdfeefghngfijefeefehkk lgmgnhnnhbpeftstefmedudkk	6.792e-11 29/40

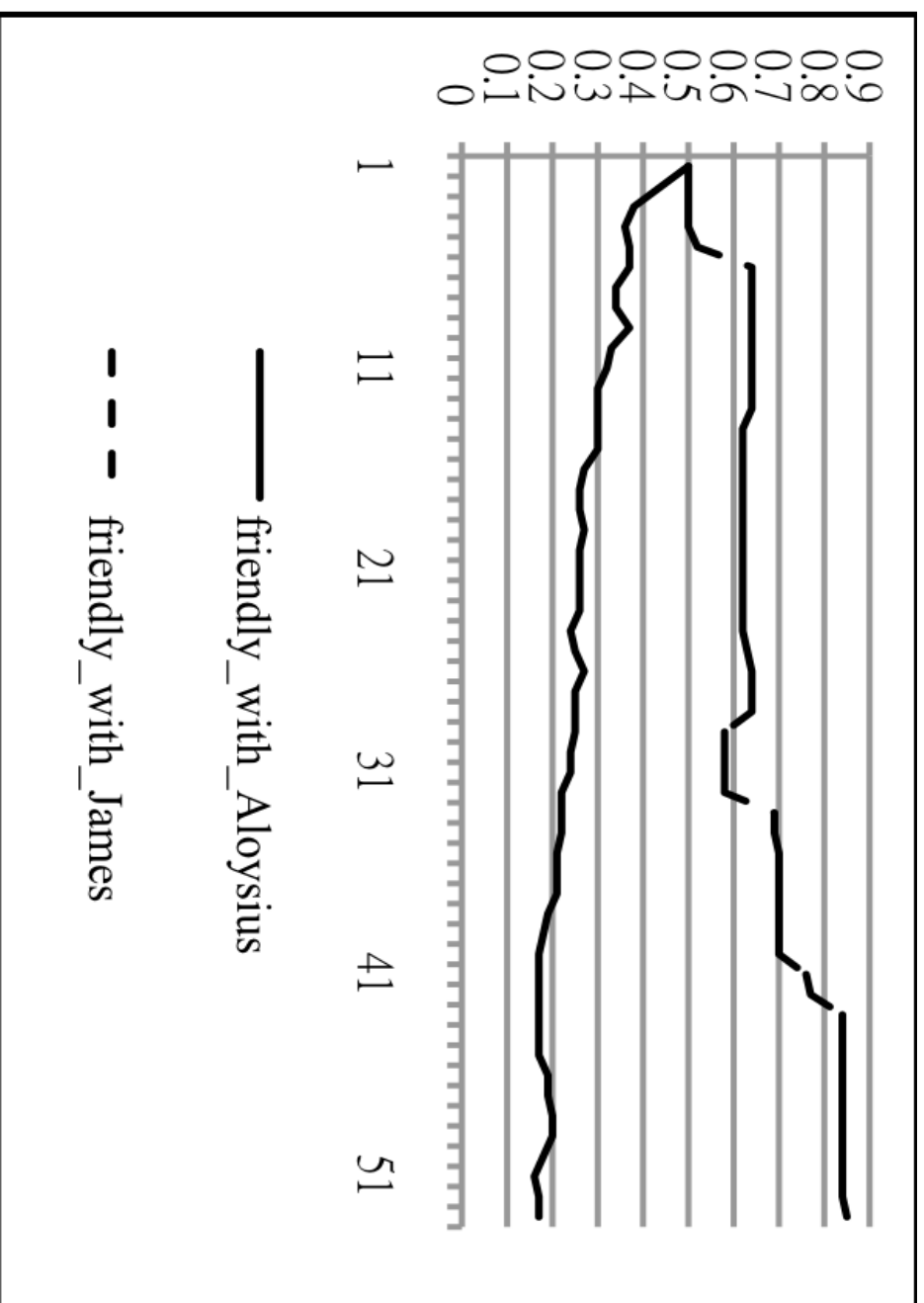
EXPERIMENT 2-2 : ESTIMATION OF AGENT'S MENTAL CONTEXTS



- We also record Father Flynn's 3 emotion states and those changes in the dialogue contexts.



INFERRING SOCIAL RELATION FROM DIALOG



CONCLUSION

- Social context-awareness requires high-level of mental models of human interactions.
- Emotion detection could be inferred to some extent from dialogue sentences if sufficient knowledge of lexical words can be properly processed.
- Speech act models seem to be feasible to bridge the gap between social context awareness with dialogues interaction
- Using Dynamic Bayesian network we would be able to link information along sequence of dialogue-acts



CONCLUSION

- Development games with virtual agents can make the story of games more flexible, and more interactive
- To make virtual agents to understand natural language as well as the story-contexts such as other agent's plan, emotion and intention, etc. and conduct dialogue freely as human seems to be itself an “AI-complete problem”?

FUTURE WORK

- The DBN requires both transition model and sensor model
- We have shown some example of using speech act as the transition model in dialogue. Currently, our sensor model has not yet been grounded to the truly signal levels
 - More social information cues such as speech tone, facial expressions and gestures obtained from “intelligent” sensors could be integrated to assist virtual agents toward social awareness.
- We could augment not only the speech act model but also augment the domain content ontology and sentence parsing and understand.



ACKNOWLEDGEMENT

- Taiwan NSC research grant
- Students: **Meng-Ju Lu, Edward Kao, Yu-Hung Chien**



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